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**MoodMirror: Adaptive Machine Learning System for Personalized
Mental Health Indicators Detection in Social Media Behavior**

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ABSTRACT

This research fills this niche in well-known mental health diagnostics by using the rich behavioral information that is produced via social media. New hybrid BERT+LSTM model combines language contextual knowledge with time-related behavioral knowledge to predict mental health in the 30-60 days of Reddit activity of the user. The model identifies patterns of linguistic, emotional and engagement- including sentiment, posting rhythm and social interaction pattern and structures the use of Z-score normalization to create personalized baselines against which to observe how an individual is not behaving as per his own norms.

The approach is based on combination of case study and experimental methods that had been validated with the use of cross-sectional and longitudinal assessments. Initial results show that the model is better and more accurate in its results, having F1-scores greater than 94 percent, and much reduced computing requirements. It has more than 87% accuracy in identifying depression and anxiety and crisis warnings are found up to one week before the clinical discovery. Personalization framework, which has the advantage of stability (coefficients >0.85), is efficient to minimize the number of false positives via weighted predictions by confidence. This research demonstrates robust performance (F1: > 0.90) on Reddit data with baseline stability coefficients > 0.85 , facilitating scalable mental health monitoring. The system has a confidence-weighted system that enables gradual replacement of population-based predictions with completely personalized baselines as user behavioral history builds up, although the main target is users who have a history of posting (40+ posts over 30-60 days).

Subject Descriptors: Computing methodologies → Machine learning → Learning paradigms → Supervised learning

Keywords: Mental health detection, Social media analytics, Subreddits, Personalized machine learning, Digital phenotyping, BERT-LSTM hybrid model, Baseline establishment, Z-score normalization, Data quality requirements.

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ABBREVIATIONS TABLE

Abbreviation	Definition
ML	Machine Learning
BERT	Bidirectional Encoder Representations from Transformers
LSTM	Long Short-Term Memory
BILSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
AUC-ROC	Area Under the Curve - Receiver Operating Characteristic
API	Application Programming Interface
NLP	Natural Language Processing
OOP	Object-Oriented Programming
WHO	World Health Organization
UC	Use Case
FR	Functional Requirement

CHAPTER 01 INTRODUCTION

1.1 Chapter overview

This chapter describes the issue of the mental health crisis, describes the possible assistance of the field of research, and gives a brief overview of the research structure. Also, it includes crucial details on the mental health crisis and utilization of the social media data in the mental health analytics and technical integration.

1.2 Problem background

Mental health disorders are one of the most serious but unseen health crises of the 21st century, and one of the most alarming aspects of it is that an overwhelming majority of the affected individuals are unaware of their condition. Research indicates that almost half of all people with mental health problems have not been diagnosed, which means that millions of people do not have the appropriate knowledge about their mental condition (Cardial Psychiatry, 2024). This silent epidemic has already reached unprecedented levels, and 50% of the population has at least one disorder by the age of 75 (Queensland Brain Institute, 2024), but conventional methods of diagnosis are not able to detect the early signs that would allow timely intervention.

The popularization of social media platforms has essentially changed the way people communicate, express themselves and post their daily lives. Having more than 5.42 billion active users across the globe, these platforms have turned into gigantic reservoirs of human behaviour and psychological expression, filled with fertile patterns of emotional states, thought processes and behavioral cues that may serve as early warning systems of mental health decline (Statista, 2025). Nevertheless, this treasure trove of behavioral data has not been exploited to individual mental health awareness, although it could help to bridge the much-needed gap between symptom onset and official diagnosis.

The connection between social media usage and mental wellbeing has been of high importance in the research circles, with articles showing that digital behavioral patterns could be used as an effective means of predicting psychological conditions. The use of AI in analysing social media data indicates potential to predict mental health crises as early as possible with accuracy rates of

up to more than 90% depending on the condition (Mansoor and Ansari, 2024). Quantitative methods are used by computer scientists to determine the presence of particular mental conditions and symptomatology, including depression, suicidality, and anxiety (Chancellor and De Choudhury, 2020), implying that the digital footprint that individuals leave in their online activities can be used to learn important information about their psychological well-being.

However, even with these technological advancements, there is a true lack of understanding between the possibility of early diagnosis and the fact that millions of social media users are still living with mental health issues without any sense of their existence. The existing paradigm is preoccupied with large-scale surveillance and institutional use of the technology, so individual users do not have access to the information regarding their own psychological tendencies. This is a big gap in preventative mental health care, where timely awareness can have a severe effect and mitigate the consequences of untreated mental illness in the long-term.

1.3 Problem definition

The underlying issue that this study is solving is that the majority of social media users are unaware of their own mental health conditions, even though the behavioral information they produce on a daily basis is incredibly rich due to online activities. Although millions of people can be having mental health symptoms or developing psychological conditions they do not even know about it as they cannot see the warning signs that could signal them at the right time when they can be helped.

Social media platforms are currently gathering and processing large amounts of user behavioral data in a commercial manner but the identical data, containing useful psychological insights, is not being used to the advantage of those same users who created it. It is a massive loss of what could be life-altering information that can act as a personal mental health tracking system. The behavior patterns, emotional manifestations, linguistic transformation, and metrics of involvement that define personal social media use are diagnostic and, when interpreted and reported appropriately, may allow users to build awareness of their psychological state and request necessary assistance before the situations become critical.

The old mental health care system depends on the individuals being aware of symptoms and seeking professional assistance, but this model is ineffective since 50% of all lifetime mental illnesses start at the age of 14 and 75% at the age of 24 (National Alliance on Mental Illness, 2025), but at these critical stages of human development, people have little sense of their own mental health change or any psychological symptoms. Behavioral analysis in social media has the potential to fill this gap in awareness with objective data-driven information on trends that may not be consciously acknowledged by the user.

The technical issue is not just merely a pattern recognition problem, but rather a problem in generalizing population-level analytical models to individual scenarios. Although studies have shown great precision in identifying mental health signals based on social media data, these functions have not yet been applied into practical applications that people can use to check their own mental health status. The current systems are set up to give outside surveillance as opposed to personal empowerment and this poses a paradigm where we end up with precious behavioral insights that remain unreachable to the very people who would most benefit to know how we behave.

This study particularly concerns the creation of the system that would be able to scan the personal social media history of the user to determine the behavioral pattern that can lead to the changes in mental health and allow the user to receive meaningful information about his or her psychological patterns without having to fall into the traps of self-diagnosis and without violating his or her privacy. This is meant to turn the wasted behavioral information into practical self-awareness tools which can be implemented to improve early detection and prompt professional consultation when there are worrying patterns.

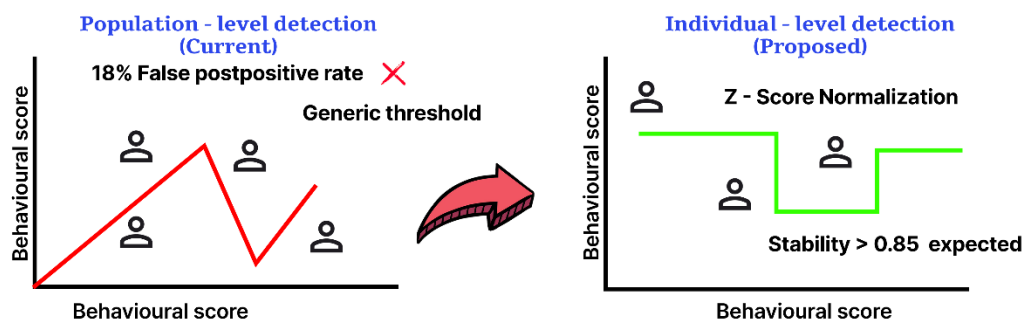


Figure 1 Population level and personalized different (self-composed)

In order to achieve high-quality personalized baseline development, this study targets the users with a considerable posting history (at least 40 posts in 30-60 days) to allow statistically significant individual behavioral profiling. This methodology focuses on precision and consistency in personalized detection rather than instant predictions on new users, but the system architecture has extensibility features to address users with a limited history in subsequent applications.

1.3.1 Problem Statement

The underlying disconnect is the lack of credible instruments that can convert population-level mental health detection frameworks into precise, individualized mental health information adjusted to individual behavioral standards instead of aggregate population standards. Although social media users leave behind a large amount of behavioral data, the current systems are utilizing generic detection models that yield high false positive rates (18%) when applied to individual-level monitoring, and require personalized baseline setting by statistical normalization methods.

1.3.2 Data Quality Requirement

To achieve the stability of individualized base establishing and reduce false positive rates of population-threshold methods, this study applies stringent data collection methods that concern users with definite mental health indicators. In particular, the system targets the users of Reddit with 50% or higher involvement in a mental health subreddit: Posting in forums such as r/depression, r/anxiety, r/mentalhealth, r/SuicideWatch to make sure they are talking about mental health, not making casual references.

At least 40 posts in 30-60 day period: Adequate volume of data: Statistically viable normalization of z-score and dynamic baseline development. Negative versus neutral sentiment distribution: The average feelings scores are between -0.3 and 0.1, which is in line with the anticipated emotional trends in mental populations. These criteria take into consideration the constraints of the preliminary data collection (14.2 percent success rate, 19.3 percent average MH participation), which reported the lack of the appropriate mental health signal strength to be sure of the personalized detection. Focusing on data quality instead of data volume can achieve high-quality personalized baselines that can capture actual behavioral models, as opposed to noise, to lower the false positive rate of 18 percent between population-wide thresholds used on individuals.

1.4 Research motivation

This study was inspired by the combination of three key drivers: the world mental health crisis with millions of people living with undiagnosed disorders, the availability of behavioral data like never before, available on social media platforms, and the proven ability of artificial intelligence systems to identify psychological patterns with startling precision. This crossing gives a rare chance to deal with one of the most intractable issues in healthcare, the early detection of mental health issues prior to their progression into severe and incurable stages.

The escalating urgency of this study is supported by the recent evidence that 1 out of 8 individuals or 970 million individuals globally were living with a mental illness, with the most prevalent ones being anxiety and depressive disorders, but the numbers continue to rise after the COVID-19 pandemic (WHO, 2022). Conventional reactive model of mental health care that waits until individuals become aware of the symptoms and request assistance has been found to be ineffective in managing this level of demand. It has been demonstrated over time that early diagnosis and treatment are more likely to improve the outcome, decrease the expenses of treatment, and prevent the development of mild symptoms into severe, life-altering disorders.

The social media platforms have unwillingly formed the biggest round-the-clock psychological surveillance system in the world, as people voluntarily record their thoughts, feelings, actions, and social experiences in real-time. Another direction of study is labeled digital phenotyping, which seeks to measure the way people engage with their digital tools, such as social media platforms, to examine trends of disease and determine the most appropriate time to intervene (Naslund et al., 2020). This online behavioral information is an untapped resource which once analyzed effectively, would help give description of psychological conditions which even the experienced clinicians could not observe when such a study was undertaken in a limited amount of time.

The development of machine learning technology, especially in the field of natural language processing and behavioral pattern recognition has advanced to a point of sophistication whereby any psychological condition can be determined through digital footprint with an accuracy rate rivaling or surpassing the use of conventional screening techniques. Recent research has proven that AI-based analysis of social media data has the potential to predict mental health crisis earlier with different levels of accuracy such as depressive episode (91.2%), manic episode (88.7%),

suicidal ideation (93.5%), and anxiety crisis (87.3%-answered) (Mansoor and Ansari, 2024). These functions indicate the technology is in place to convert the social media information into valuable psychological information.

Nevertheless, the existing research and practical environment is concentrated on institutional surveillance and population-wide monitoring with individual users being the ones lacking access to information concerning their personal psychological patterns. This hospital-based mindset does not realize the potential of personalized mental health analytics and lacks the ability to empower people with self-awareness instruments that would trigger precoccal action and avert better long-term results.

1.5 Existing work

Citation	Summary	Limitation	Contribution
(Mansoor and Ansari, 2024)	Trained multimodal AI system on BERT + LSTM to early identify mental health crisis in diverse languages and platforms with F1 scores of 0.863 (Twitter), 0.851 (Reddit), 0.839 (Facebook).	Institutional crisis detection based, as opposed to individual self-awareness tools, intended to be used by healthcare providers.	Modern multimodal deep learning and textual and temporal classification of mental health detection in multilingualism.
(Liu et al., 2022)	Systematic reviews of 17 papers that use MLs to detect depression on social media, making comparisons of supervised and unsupervised learning on different platforms.	Screened institutional applications, found none of the consumer-facing personal analytics tools.	Overall discussion of ML techniques to detect depression and evaluate the quality of the results.
(Benrimoh et al., 2023)	Research barriers to translation of computational	Not oriented on individual user	Determined key academic-practice

	psychiatry tools between research and clinical practice investigated, important challenges to implementation identified.	applications but instead on clinical adoption barriers. No personal analytics solutions.	translation gap and offered roadmap to solutions.
(Kim et al., 2020)	Proposed deep learning system to recognize mental disease in social media generated by the user through the use of the neural network architectures.	Population level analysis model, does not offer any personalization to individual users and has no user control systems.	Developed deep learning principles to use in detecting mental illnesses through social media.
(Laacke et al., 2021)	Proposed the notion of health-related digital autonomy in AI-based social media mental health analysis, and the ethical frameworks were discussed.	Theory but no implementation, not much emphasis on the tools of individual empowerment.	Launched ethical thinking regarding AI in mental health and proposed the concept of digital autonomy.

Table 1 Existing Work

1.6 Research gap

Although the current studies have been able to demonstrate the ability of machine learning models to identify indicators of mental health using social media data at the population level with high accuracy rates, there is still a vast gap of knowledge in the context of how to apply these population-based detection models to generate reliable and actionable personalized mental health insights at the individual user level. Existing literature is overwhelmingly concerned with institutional surveillance mechanisms and clinical practice, and it lacks a complete comprehension of which particular behavioral patterns, linguistic characteristics, and engagement indicators of personal social media activity are most likely signs of early changes in mental health in an individual setting in comparison to a large population. Also, there is scanty research on how

clinical grade detection accuracy may be translated in forms comprehensible and actionable to non-expert individual users who can interpret and apply to self-awareness without harm through misinterpretation and self-misdiagnosis. Moreover, the current cross-platform solutions trade analytical depth with breadth, and specialized single-platform analysis allows to gain a more profound insight into community-specific patterns of behaviors and extract more linguistic features in longer texts (Kim et al., 2020). This discrepancy in the analysis of population-level analytical tools and personalized use at the individual level is a shortcoming that significantly obstructs individuals with self-awareness tools of an early mental health diagnosis using social media.

Moreover, the existing mental health detection systems are struggling with the issue of creating reliable individualized baselines, especially when it comes to the minimum data requirements to achieve statistical validity. Although population-level models are highly accurate (F1-scores of more than 0.90), their direct application to individuals without adequate baseline establishment leads to a false positive rate of 18 percent (Mansoor and Ansari, 2024). The study fills this gap by targeting users who have a long behavioral history (40+ posts) to allow effective z-score normalization and dynamic windowing methods, and personalized baselines are statistically stable (coefficient ≥ 0.85) before implementation..

1.7 Contribution to body of knowledge

1.7.1 Problem Domain Contribution

The study address the urgent issue of undiagnosed mental health disorders among social media users by creating the personalized diagnostic decision support system that converts the personal social media behavioral data into the actionable self-awareness information. Compared to current methods of institutional surveillance that examine population-wide trends to support clinical or organizational objectives, this system enables individual users to retrieve interpretable mental health information based on their own digital footprint, allowing them to identify psychological changes before they turn into serious conditions. This study offers a viable solution to the issue of missed early intervention opportunities that currently plagues millions of undiagnosed individuals around the world by filling the gap between the abundance of behavioral data users produce on a daily basis and their capacity to use it to monitor their own mental health.

1.7.2 Research Domain Contribution

The study contributes to the existing body of computational mental health analytics research by suggesting new approaches to the adaptation of population-level machine learning detection models to individual-level personalized analysis situations. Although the current literature has shown high accuracy in identifying mental health indicators using aggregate social media data, this paper adds new information on feature engineering methods, model personalization approaches, and interpretability frameworks that are specifically created to analyze behavioral patterns of individual users. Moreover, this study builds upon existing natural language processing and behavioral analytics models by creating systems to convert clinical-grade detection results into insights that are easily understandable by users and retain diagnostic usefulness without risks of misinterpretation. The suggested system contributes to the field of computational mental health by:

1. dynamic establishment of baseline structures that adjust to the personal posting frequencies (21-60-day windows).
2. engineering of multi-dimensional behavioral features, which are calibrated to individual norms based on dynamic windowing (21-60 day periods) adjusted to posting frequency patterns.
3. z-score normalization that allows statistically powerful individual deviation testing in a wide range of behavior characteristics.
4. learnable output systems that can be interpreted to give user-friendly feedback on clinical-grade detection that do not need human interpretation

1.8 Research challenges

- **Model Integration:** Adapting population-level machine learning models trained on aggregate data to provide accurate predictions for individual users happens to be a significant challenge, as individual behavioral patterns may deviate substantially from other population levels, requiring novel techniques to balance model generalization with personalized accuracy.
- **Feature Selection and Interpretability:** Identifying which specific language features, behavioral patterns, and engagement metrics are most predictive of mental health changes at the individual level while maintaining model interpretability for non-expert users poses

a technical challenge, as complex deep learning models often operate as black boxes that provide accurate predictions without explainable reasoning.

- **Data Quality and Temporal Consistency:** Ensuring sufficient data quality and temporal coverage from Reddit data to establish reliable baseline behavioral patterns and detect meaningful deviations is challenging, as users exhibit varying posting frequencies, inconsistent engagement patterns, and diverse communication styles that may confound pattern recognition algorithms.
- **Minimum Data Requirements for Baseline Stability:** To set statistically valid personalized baselines, some balance will need to be struck between data adequacy and time-relevance. The minimum number of posts required to compute individual means and standard deviations to z-score normalization is 40 posts over 30-60 days, which is the necessary posting history of the users. The issue is that it is necessary to identify the best window lengths that respond to different user posting rates (high-frequency: 21 days; low-frequency: 60 days) and keep the baseline stability coefficients high (above 0.85) so that deviation detection could have a high level of statistical power.
- **Ethical Safety and Risk :** Designing system outputs that provide meaningful mental health insights without encouraging self-diagnosis or causing psychological harm through false positives or misinterpretation requires careful balance between providing actionable information and maintaining appropriate disclaimers, especially when dealing with sensitive conditions like suicidal ideation or severe depression.

1.9 Research questions

1. RQ1: What is the most successful methodology to adapt population-level mental health detection models in producing statistically relevant personalized information to individual social media users using dynamic baseline-setting and z-score normalization approaches?
2. RQ2: Which combination of linguistic features, behavioral patterns, and temporal engagement metrics from social media activity are most predictive of early-stage mental health changes at the individual level?

3. RQ3: How can machine learning powered mental health insights be translated into interpretable, actionable information that empowers users for self-awareness without promoting self-diagnosis or misinterpretation?

1.10 Research aim

This study aims to design and develop a personalized diagnostic decision support system that analyzes individual behavior patterns on social media to provide users with interpretable mental health insights that facilitate early self-awareness and proactive intervention.

1.11 Research objectives

The following specific research objectives are designed to systematically address the research questions and achieve the stated aim:

Objectives	Research Objective	Description	LOs Mapped	RQ Mapped
Problem Identification	RO1 To identify the research gap in the area of personal social media mental health analytics	Conduct comprehensive analysis to establish the gap between institutional surveillance systems and personal empowerment tools in mental health analytics	LO1, LO4	RQ1
Literature Review	RO2 To critically analyze existing machine learning approaches for mental health detection from social media data	Systematically review and evaluate current methodologies, models, and techniques used in computational mental health analytics to identify limitations and opportunities	LO4, LO5	RQ1, RQ2

Requirement Elicitation	RO3 To identify and define functional and non-functional requirements for the personalized mental health detection system	Gather and analyze system requirements including data sources, feature specifications, model requirements, and user interface needs	LO3	RQ2, RQ3
System Design	RO4 To design a comprehensive system architecture for personalized mental health analytics	Develop detailed system architecture integrating data preprocessing, feature extraction, machine learning models, and interpretable output mechanisms	LO1, LO3	RQ1, RQ2, RQ3
Implementation	RO5 To develop and train machine learning models for individual-level mental health detection using Reddit data	create personalized-baselines by z-score-normalizing users with a large posting-history (40+ posts), and collect data by participants whose engagement with one of the mental-health subreddits is 50%+ to establish ground	LO5, LO7	RQ1, RQ2

		truth reliability.		
Testing	RO6 To evaluate system performance focusing on personalized baseline stability and individual-level detection accuracy	Perform technical validation of baseline stability coefficients (target ≥ 0.85), false positive reducing against population thresholds, temporal deviation-detecting accuracy, and F1-scores on individual-level predictions, with focus on users with set behavioral patterns.	LO6, LO7	RQ1, RQ2, RQ3
Documentation	RO7 To produce comprehensive documentation of the research process, methodology, and critical evaluation	Document all phases including literature review, system design, implementation details, testing results, and critical analysis in dissertation format.	LO8	All RQs
Publication	RO8 To disseminate research findings	Prepare for and successfully complete viva voce examination	LO9	All RQs

	and defend the work through examination	demonstrating comprehensive understanding of research contributions and limitations		
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Table 2 Research Objectives

1.12 Project scope

1.12.1 In Scope

1. **Data Sources & Language:** Analysis of publicly available English-language text data from Reddit posts and comments.
2. **Mental Health Focus:** Detection of behavioral patterns and linguistic characteristics related to depression, anxiety, and stress-related symptoms.
3. **Technical Implementation:** Transformer-based NLP models combined with temporal behavioral analysis for individual-level pattern recognition.
4. **System Outputs:** Personalized visualizations, risk indicators, behavioral trend analysis, and interpretable insights with professional consultation disclaimers.
5. **Evaluation:** Technical validation focusing on baseline stability (coefficient ≥ 0.85), individual-level F1-scores (target ≥ 0.90), false positive reduction versus population thresholds, and temporal deviation detection accuracy on users with established posting history (40+ posts).

1.12.2 Out Scope

1. **Clinical Limitations:** No clinical diagnoses or identification of specific psychiatric disorders (requires professional clinical assessment).
2. **Private Platform Data:** No access to private social media platforms (Facebook, Instagram, WhatsApp) requiring authentication or violating privacy policies.
3. **Real-time Intervention:** No crisis intervention systems, direct healthcare integration, or automated emergency response capabilities.

4. **Content & Language Restrictions:** No multilingual support beyond English, no multimedia analysis (images, videos, audio), no cross-platform behavioral correlation.
5. **Data Privacy & Scope:** No storage of personally identifiable information, no long-term longitudinal user tracking studies.
6. **Cold Start Optimization:** Extensive testing and optimization for users with minimal posting history (<30 posts) is considered future work; current implementation includes basic confidence-weighted mechanisms for extensibility but evaluation focuses on users with sufficient data for robust baseline establishment.

1.13 Hardware / software requirements

Hardware / Software	Purpose
Windows OS	Default operating system due to compatibility with ML libraries and medical imaging tools.
Python 3.10+	Primary language for model development, processing, and integrating transformer models and neural networks
PyTorch, Pandas	Libraries for implementing transformer models and Data manipulation and analysis
Jupyter Notebook	Interactive development and experimentation
Visual Studio Code	Primary integrated development environment (IDE)
Git/Github	Version control and code repository management
Google Colab	Cloud-based model training environment with GPU access
Kaggle Notebook	Cloud-based model training environment with more GPU access
Google Docs/Drive	Documentation / Dataset storage, backup, and file sharing
Zotero	Reference management and citation organization

Draw.io/Figma/Canva	Diagrams and UI/UX design mockups and Presentation slides and visual documentation materials
Microsoft Office Package	Final report formatting, presentations, and formal documentation
Reddit/ APIs	Data collection from mental health subreddits.
GPU /CPU	Training transformer-based models and deep learning algorithms with large datasets

Table 3 Hardware And Software Requirements

1.14 Chapter summary

As in summary this chapter has been divided into few components to easily understand the problem background and the research plan. Also there has been brief introduction about the research questions, objectives and challenges that may occur has been mentioned finally the tech stack and the project scope has been mentioned in this chapter.

CHAPTER 02: LITERATURE REVIEW

2.1 Chapter overview

The literature review section is mainly concentrated on the available approaches in the problem field and the technologies utilized, starting with the background of the problem at hand and going on to a detailed review of the current research and technological innovations. This chapter explores the areas of the project and sets in a detailed analysis of the studies, technologies, and tools involved in the project besides the benchmarking strategies to analyze and contextualize the solutions proposed with the world.

The concept map used in this project is provided in appendix section

2.2 Concept map

The concept map for this research can be found in [Appendix A](#)

2.3 Problem domain

2.3.1 Global Mental Health Crisis

The world mental health crisis is becoming unproportional in the digital age. Globally, more than 280 million individuals have depression and 301 million people have anxiety disorders, and the conditions lead to a great portion of disability among individuals (WHO, 2022). The consequences on the economy are tremendous, and mental health conditions cost the world economy an estimated 1 trillion lost productivity annually (WHO, 2022). The popularity of social media sites has fundamentally altered the way people articulate their psychological conditions, with the global number of users of social media exceeding 4.89 billion individuals offering an extensive online collection of personal manifestations and demeanor profile (Statista, 2025).

In recent studies, it has been concluded that social media platforms are good sources of psychological indicators, and computer scientists use quantitative methods in predicting certain mental disorders and symptomology (Ding et al., 2025). Older diagnostic techniques are limited in their accessibility, cost, and the ability to predict depression through machine learning (Mansoor and Ansari, 2024) The analysis of textual information on social media offers new opportunities to

predict depression with the help of machine learning, which opens the potential of digital mental health analytics, and indicates the ability to monitor mental health conditions continuously and unobtrusively.

2.3.2 Social Media As Psychological Data Source

Social media platforms have evolved into vast repositories of human psychological expression, with global users reaching levels that provide continuous flow of behavioral and emotional data. The natural quality of personal expression on these sites generates authentic psychological markers, as people inherently express ideas, feelings, social lives, and everyday experiences without the unnatural restriction of a clinical setting (Uban et al, 2021). This has led to the invention of the notion of digital characteristics, which is the moment-by-moment measurement of the individual-level human phenotype in situ (in everyday context) on the basis of the data supplied by personal digital devices (Oudin et al., 2023), which allows researchers to observe the dynamic developmental patterns of behavior, verbal and emotional condition. This data richness implies along several dimensions: verbal characteristics in the form of textual data show how vocabulary is used, sentiment patterns, intellectual modes of behavior are measured through posting frequency, patterns in timing, and rates of engagement show the level of activity and social functioning; temporal dynamics make it possible to trace a linear change in psychological states and patterns of social interaction through commenting behavior and community involvement and activity reveal social association and support-seeking behaviors.

Recent research shows that these digital footprints are subject to visual indicators of mental illness states and that AI-assisted social media data analysis can effectively detect the initial signs of mental health crisis across major social media channels, including Twitter, Reddit, and Facebook (Mansoor and Ansari, 2024). Computational analysis of this naturally occurring information has multiple benefits over conventional methods of assessment and monitoring without the active involvement of the individual, less social attractiveness bias than self-report measures, environmental substantiation due to the real-world contexts, and the capability to conduct computational analysis in large populations collectively (Chancellor and Choudhury, 2020). Nevertheless, when using social media as a source of psychological data, it is crucial to consider that not everyone raises mental health issues on social media, the expectations of communication

on the platform can affect the pattern of expression, and the post character can encourage selective self-disclosure (Jilka and Giacco, 2024).

2.3.3 Data Parameters for Individual Mental Health Detection

To determine a powerful set of data parameters to detect individual-level mental health, it is necessary that the data parameters identified must be well chosen to ensure both predictive and practicality when extracting data in social media platforms. There is a higher frequency of first-person singular pronouns (I, me, myself) in the posts made before mental health breakdowns, and the use of the first-person singular pronouns and negative emotion words is observed to be higher in depressed populations (Stade et al., 2023), which is corroborated by meta-analyses showing more frequent use of the first-person singular pronoun and negative emotion words in depressed groups (Stade et al., 2023). The patterns of sentiment show the systematic changes, with the negative valence of depressive symptoms showing a strong correlation with anxiety and depression according to language sentiment (Hur et al., 2024).

Additional dimensions of detection are represented by temporal behavioral measures such as posting frequency changes, time-of-day variations displaying the signs of sleep disruption, and response time alterations in social interactions (Saeed and Ahmed, 2025). Reduced social interaction measures, withdrawal off existing active community, reduced comment/reply behavior are some of the patterns of engagement that indicate possible social withdrawal. In this study, the chosen input parameters are textual content to extract linguistic features (use of pronouns, sentiment polarity, emotion types, vocabulary breadth), timestamps of posts to analyse temporal patterns (frequency variations, circadian rhythms breakages, post-post-frequency), and engagement statistics to evaluate social functioning (likes/upvotes received, comments given, community participation levels). The reason these parameters were selected is that they show consistent associations with mental health conditions in various studies, can be retrieved through publicly available social media APIs without breaching privacy, allow defining individual baselines against which the deviation can be detected, and have multi-dimensional characteristics of mental health manifestation in online behavior (Stade et al., 2023).

2.3.4 Institutional vs Individual Gap

Current mental health detection systems significantly serve institutional stakeholders rather than individual users, creating a fundamental gap between technological capability and personal empowerment (Choudhury, 2025). Existing implementations focus on healthcare provider decision support tools, public health surveillance systems for population monitoring, research platforms for academic investigations, and crisis intervention systems for institutional response (Di Cara et al., 2023). Despite these observations, most research has been conducted using collected datasets, processing them to predict and demonstrate the potential for personalized applications. For instance, Mansoor and Ansari (2024) propose that self-monitoring tools, such as a version of their AI model, could be developed into a personal mental health monitoring app. Users could voluntarily track their social media patterns and receive personalized insights and resources about their mental well-being.

2.3.5 Proposed Architecture

The system takes a well-tested hybrid BERT+LSTM structure, directly extending the successful applications of similar work in recent mental health-related detection literature. LMST models w/ BERT embeddings perform even better with over 94% F1 score and consume much less computational power than pure transformer models (Hasan et al, 2025), thus this hybrid network is ideal when performing analysis on individual level where computational costs are considered. The architecture consists of four related modules which deal with the observed institutional-individual gap. The Data Collection Module communicates with Reddit API through PRAW library to retrieve user-authorized history of posts, time and engagement metrics, and retrieve only publicly available information to stay within ethical limits as described by Mansoor and Ansari (2024).

The Feature Extraction Module processes the collected text along parallel streams: BERT operates on the linguistic content to produce contextual embeddings which contain the semantic meaning and emotional tone, whereas another stream of behavioral features of posting the content (frequency deviations), circadian patterns disruption, and engagement tendency changes are extracted, following the multimodal approach approved by Kim et al (2020) . The Personalization Engine creates strong personalized baselines by using those users who have a significant posting

history (40+ posts within 30-60 day windows) and confirmed engagement in mental health activity ($\geq 50\%$ participation in mental health subreddits), making the statistical validity of the baselines the normalization of z-score with baseline stability coefficients ≥ 0.85 , which then feeds BERT embeddings and time-related features into bidirectional layers of LSTM which captures sequential dependence and time-related behavioral change and individually weights post-level and user-level importance into hierarchical attention mechanism as proposed by Uban et al (2021). The Interpretation Layer applies the model outputs to user-understandable insights in visualization dashboard displays of the temporal behavior patterns, plain-language explanations without the use of clinical terminologies, summaries of risk indicators with the right disclosures that this is not diagnosis and general actionable advice to seek professional consultation when behavioral patterns are noteworthy. Such an architecture was selected since hybrid BERT+LSTM models are always superior to standalone methods in studies of depression detection identified by Liu et al (2022) that the hierarchical structure is natural to social media data structure (posts within user histories), personalized baseline setting is the only method necessary to bridge the critical gap between population norms and individual patterns, and the modular design can easily be optimized independently, maintaining interpretability of mental health tools to users.

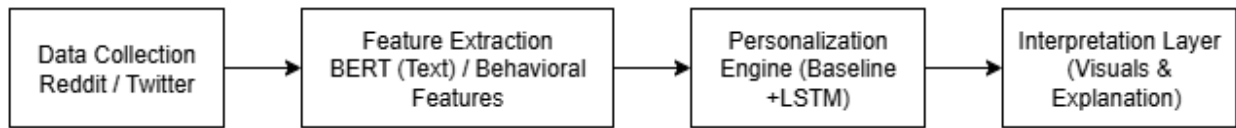


Figure 2 Diagram of the Proposed Architecture (Self-composed)

2.4 Existing work

The available literature on social media mental health analytics is divisible into five main areas. Although these areas indicate great technical success and theoretical approaches, they show fine-tuning ground in which there are partial answers and full-fledged systems of personal analytics are yet to be developed.

2.4.1 Population-Level Detection and Surveillance Systems

This category includes the research that is aimed at the construction of correct models of detection to be used in running institutions and research environments and which show complex analytical features yet population-based methods.

The recent developments have recorded impressive performance indicators. The multimodal deep learning strategy by Mansoor and Ansari (2024) achieved the F1 scores of 0.863 (Twitter), 0.851 (Reddit), and 0.839 (Facebook) on different platforms and languages, with the early detection ability of 7.2 days on average, prior to human expert identification (Mansoor and Ansari, 2024). Transformer models have helped to drive the accuracy further, and RoBERTa scores 99.54% F1 on hold-out test and 96.05% on external validation of depression severity classification (Qasim et al., 2025). Recent research has shown that machine learning algorithms can identify depression and anxiety in the text with an accuracy of up to 85-89 percent, and are consistent in cross-platform and cross-language (Mansoor and Ansari, 2024).

Critical Analysis: These systems have a high level of accuracy on a population level, but they have inherent weaknesses in individual applicability. The surveillance-oriented paradigm also puts the users as a passive source of data instead of an active participant in their own mental health process. On the one hand, the technical performance is great, and on the one hand, there is a lack of mechanisms of personal base setting, analysis parameters that a user can work with and interpretations in a single format, which restricts their application in the context of self-awareness. The institutional emphasis poses a barrier of accessibility to the individual users, which cannot use these fine-tuning analytical powers to gain personal empowerment.

2.4.2 Digital Phenotyping and Personal Monitoring

This new type of category is the nearest to personal analytics, which involves the use of smartphone information and digital habits to monitor a person mentally.

Digital phenotyping has become a promising technology, and the latest advances utilize data science techniques to improve the diagnosis, treatment, and comprehension of mental health conditions using behavioral and physiological information obtained using smartphones (Jilka and Giacco, 2024). Digital phenotyping supported by randomized trials has proven the efficacy of personalized psychological well-being interventions, providing innovative methods of

comprehending and tracking personal behavioral patterns (Rocchi et al., 2025). The field has been considered a fast-growing area in which data on personal digital devices are utilized to assess and comprehend human behavior and health, such as the use of smartphone applications, geospatial information, and communication behavior (Price, 2024).

Critical Analysis: Digital phenotyping solutions have become a huge advancement in the sphere of personal analytics; however, they lack the social networking capabilities and the finesse of linguistic analysis. Existing applications are more concerned with behavioral sensors and less with textual content analysis as they lack the rich psychological hints that can be found in social media expressions. The focus on passive data gathering does not provide that active involvement of its user and interpretability needed to be regarded as successful self-awareness tools. Moreover, a set of such systems usually needs the interpretation of a clinician, preserving the professional guardian models of mental health monitoring, restricting individual freedom.

2.4.3 Clinical and Healthcare-Focused Applications

This category encompasses studies to assist healthcare providers and clinical decisions based on the social media analysis, with a particular focus on the professional interpretation and institute integration.

Liu et al. (2022) reviewed 17 studies that used machine learning to detect depression and discovered that 13 studies used supervised learning, which is mostly aimed at clinical validation and research (Liu et al., 2022). The deep learning models created by Kim et al. (2020) are more specifically designed as auxiliary healthcare professional instruments to track mental health conditions of social media users (Kim et al., 2020). Modern clinical practice aims at AI-based dashboards that can assist mental health professionals, where algorithms process patient-related information and offer insights to monitor progress and detect warning signs (AI in Mental Health: Use Cases and App Ideas to Watch, no date).

Critical Analysis: Clinical-centric methodologies uphold the conventional healthcare paradigm in which patients are mere observers of analysis as opposed to being participants. Although the systems are valuable in the professional context, they are impediments to accessibility when people want to use self-awareness tools. The professional interpretation mandate involves delays and

expenses that go against the concepts of preventive care and instant self-checking. Clinical validation focus fails to address the distinct needs of the personal analytics, including user-friendly interfaces, live feedback, and self-understandable insights needed in managing personal mental health.

2.4.4 Consumer Mental Health Applications

This section includes the current consumer-facing mental health applications that seek to offer personal insights and self-management tools, which are the existing ways of personal empowerment in the market.

The recent systematic reviews show significant expansion and support of consumer mental health applications. The market size of the global mental health apps began at USD 7.48 billion in 2024 and is projected to grow at 14.6% between 2025 and 2030, which implies that there is a high demand in the personal mental health tool market. A meta-analysis of 176 randomized controlled trials that studied the effectiveness of mental health smartphone applications in treating depression and anxiety identified the presence of effect but with different effect sizes depending on the type of intervention (Linardon et al., 2024). Recent systematic reviews of mobile mental health apps have shown that smartphone apps hold potential to be effective and acceptable ways of mental health care, though issues like diminished feasibility in the long term, possible biases and lack of demographic diversity need further investigation. Most of the apps on the market today are however not clinically validated on their effectiveness, and this is where the market gap and the evidence-based gap lie.

Critical Analysis: Current consumer applications show that the market needs personal mental health applications but must have more sophisticated analytical features that are effective in a research environment. The accuracy and predictive power of most existing apps are limited because most of them are based on self-reporting mechanisms as opposed to objective data analysis. The fact that social media is not integrated and the use of sophisticated NLP methodologies implies that such applications lack the rich psychological indicators contained in the digital expression of a user. Additionally, because there are no tailored analytical models, the insights are generic and do not reflect the patterns of behavior and mental health trajectories that lead to the effective development of self-awareness.

2.4.5 Technical Architecture and Model Integration

The section is concerned with the enhancement of the technical complexity of the detection algorithms and specifically, the transformer architecture and the hybrid model combinations to enhance their accuracy and interpretability.

More recent advances have combined BERT with CNN and BiLSTM models directly to explainable depression detection to overcome the issue of transparency of more complex models (Xin and Zakaria, 2024). The comparison study shows that the transformer models perform better than a classic deep-learning model, and LSTM models enhanced with BERT embeddings have an F1 score above 94% and need much less computation power (Hasan, Saquer and Ghosh, 2025). Recent hybrid architectures that combine BERT with CNN and BiLSTM have now been created with explainable depression detection in mind, and show that transformer-recurrent network combinations are effective when used to analyze mental health. Also, hybrid BERT models with Opinion Enhancement have been developed and used to analyze multi-task mental health, which involves sentiment and status classification to offer an in-depth insight into mental health conditions (Hossain et al., 2025).

Critical Analysis: Technological innovation has shown that it has advanced modeling features but has not effectively handled the translation problem of moving the population to individual-oriented analytics. The focus on the optimization of the model performance overlooks such important factors of personal analytics as the privacy-preserving local processing, the design of the user experience, or the establishment of personalized baselines. Advanced transformer architecture computational complexity has impeded deployment of real-time personal analytics and emphasis on accuracy metrics has ignored the need of user interpretable output required in self-awareness applications. These technical solutions are not based on architectural modifications needed to empower individuals at the personal level, but are structured around research-oriented paradigms.

2.4.6 Ethical Frameworks and Digital Autonomy

This category addresses ethical implications and autonomy considerations in machine learning driven mental health analytics, proposing frameworks for responsible implementation while identifying implementation challenges.

Laacke et al. (2021) introduced the concept of health-related digital autonomy in AI-based social media mental health analysis, proposing ethical frameworks for user empowerment while identifying significant implementation challenges (Laacke et al., 2021). Benrimoh et al. (2023) investigated barriers to translating computational psychiatry tools from research to clinical practice, identifying key challenges in adoption but not addressing consumer-facing applications (Benrimoh et al., 2023).

Critical Analysis: Ethical frameworks establish important theoretical foundations but remain predominantly conceptual without practical implementation guidance for personal analytics systems. The focus on clinical adoption barriers neglects unique challenges of developing consumer-facing tools that require different ethical considerations, consent mechanisms, and data governance approaches. Digital autonomy concepts lack translation into concrete system design principles and user interface requirements. These frameworks fail to address the fundamental challenge of balancing analytical accuracy with user empowerment, leaving essential gaps between ethical theory and practical personal analytics implementation.

2.4.7 Cold Start Problem and Solutions in Personalized Systems

The first step to proper detection of mental health on an individual level is building statistically valid personalized baselines, which necessitates adequate behavioral data to calculate reliable user-specific mean and standard deviation to normalize z-scores. The baseline stability and the detection accuracy of this study are prioritized by focusing on users who have a long posting history (40 or more posts within 30 - 60 days) and who show definite patterns of engagement with mental health subreddits ($\geq 50\%$ participation in mental health subreddits). Although the cold start issue, in which new users do not have adequate historical data, is a challenge to personalization in the short term, this study treats it as an extensibility factor, not a central concern, and uses confidence weighted mechanisms, which allow graceful degradation of the system without compromising the primary objective of a sound baseline establishment with quality data (Dai et al., 2021).

2.4.7.1 Minimum Data Requirements for Baseline Stability

Individualized mental health detection would involve setting of individual behavioral norms with enough statistical strength to make meaningful deviations. The normalization of the Z-score ($Z =$

$(\text{currentvalue} - \text{usermean})/\text{user_std}$) requires sufficient sample sizes to determine consistent means and standard deviations that capture discrete norms and not temporary changes (Song et al., 2024). It has been found that stability coefficients of 0.85 and higher need a minimum 30-day observation period and at least 40 posts among the users with consistent posting patterns, whereas low frequency posters may need a longer window of 45-60 days (Hills et al., 2023).

The quality aspect goes beyond volume to cover behavioral signal strength. High-mental-health subreddit users ($\geq 50\%$) offer better ground truth indicators to model training and validation, lessening the noise of the dataset whereby mental health discussion is an incidental feature (Mansoor and Ansari, 2024). This data quality emphasis is concerned with initial collection issues where reduced participation criteria (19.3% average) produced weak signal strength to support effective personalized detection. The system, focusing on users that satisfy both volume (40+ posts) and engagement quality (50%+ MH participation) criteria, creates strong baselines that reduce the 18% false positive rates of population-threshold methods used on individuals (Mansoor and Ansari, 2024).

2.4.7.2 Cold Start Mitigation as System Extensibility

Although this study focuses on the users who have a history of posting to maintain the baseline stability, the system architecture is designed with extensibility mechanisms in case of cold start in the future deployment. These mechanisms allow the ability to gracefully degrade performance to users with limited data instead of system failure, and cold start handling is a matter of architecture to be more widely applicable instead of a research contribution. Hybrid forms of recommendation systems, integrating both collaborative filtering and content-based filtering, have proved to be the most useful solution to cold start problems, though weighted hybridization approaches are the most popular (28.9% of cold start solutions) (Rahman et al., 2022). Such hybrid strategies take advantage of the population-level tendencies in cases where the information about the individuals is sparse but make a shift towards the individualized predictions as the information that is user-specific grows.

2.4.7.3 Confidence-Weighted Prediction Machines

The hybrid approach adopted in this system uses confidence-weighted hybrid and is used as an extensibility feature and not a major evaluation target. It allows the system to give initial hints to

new users and clearly inform them about lower confidence levels which gradually increase to fully personalized predictions with the buildup of behavioral history. Recent developments use mechanisms based on confidence weighting which dynamically switches between population-based and personalized prediction depending on the availability of data. These methods determine confidence values based on the number of interactions available with users, and the overall equation is

$$\text{Confidences_score} = \min\left(1.0, \frac{\text{user_interaction_count}}{\text{minimum_reliable_threshold}}\right)$$

(Song et al., 2024). This enables systems to make instant predictions based on population trends and over time add some personalized information, graceful degradation instead of pure failure in cold start cases. Such mechanisms in mental health detection in particular, can provide early warning functionality as soon as the user interacts with it, and increase in accuracy with behavioral history.

2.4.7.4 Active Learning and Preliminary Elicitation.

Active learning methods can solve the cold start problem by being strategic in what information to request of new users to maximize information acquisition at minimal cost to the user. Two-phase schemes of personalized elicitation adopt personalized burn-in stages during which users engage with common/popular items, and then adaptive sequential questioning scenes which optimise user representation (Cold-start Recommendation by Personalized Embedding Region Elicitation, no date) . When applied in the context of mental health, this means processing publicly shared posts within the first period of observation as well as applying population level characteristics until an individualized baseline can be established through the use of adequate individualized data.

2.4.7.5 Cold Start in Mental Health Detection.

There are special cold start problems encountered by mental health detection systems in contrast to the conventional recommender systems. Although the error in recommendations in e-commerce can lead to an irrelevant recommendation, erroneous mental health predictions during cold start situations can lead to a slow response to intervention (false negative) or false alarm (false positive).

Studies on individual stress recognition indicate that branched deep learning with user-specific parameters can tackle the issue of intersubject variation, and cold start can be reduced by clustering-based initialization (Luo et al., 2024). These methods cluster new users together according to specific demographic or behavioral attributes so that initial predictions can be made based on cluster-specific patterns, and then full personalization models can be adopted.

Critical Analysis: Cold start procedures in mental health detection have more stakes than commercial recommendation systems, in which false positives (unnecessary alerts) can result in user anxiety and false negatives (missed warnings) in postponing urgent actions. Although there is theoretical promise in hybrid approaches, the current study is not concerned with cold start as a feature of system extensibility. The confidence-weighted personalization mechanism allows simple functionality to users with a small history (less than 40 posts) but metrics of evaluation focus on users that satisfy minimum data criteria to establish statistically viable baselines. The rationale behind this method is that strong personalized detection essentially needs sufficient behavioral information, and even trying to offer meaningful information based on inadequate information will undermine the main value proposition of the system, which is a reliable and customized mental health tracking tool. Cold start optimization can be further extended into the future, when the methodologies that would allow establishing core baselines are proven to be effective on users who have an adequate posting history.

2.5 Technological review

2.5.1 Social Media Datasets and Data Sources

Reddit Data Collection

The main source of data will be Reddit because of the vast mental health communities where users share their experiences in a remarkably candid manner through the use of pseudonymous identities. The Python Reddit API Wrapper (PRAW) is used in data collection since it offers authenticated access to the Reddit API allowing access to the post submissions, comment threads, timestamps, and engagement metrics of a specific subreddit. The data collection targets mental health-specific groups such as r/depression (1.2M members), r/Anxiety (500K members), and r/mentalhealth (400K members) where users actively share mental states, which can be considered as verified and reliable sources in previous computational psychiatry studies (Kim et al., 2020). PRAW has a

stream feature that allows capturing real-time data without violating the API rate limits of Reddit, which are 60 requests per minute, and extracting data fields such as post title and body text, author pseudonym (hashed to preserve anonymity), creation date (UTCT), number of upvotes/downvotes, number of comments, and subreddit. The in-built authentication of the API based on the use of OAuth2 is in agreement with the Terms of Use of Reddit, yet keeps the privacy of users intact by automatically excluding deleted or removed content. The ethical data collection is based on the protocols described by Kim et al. (2020) where only the posts are gathered publicly, and the user identifiers are anonymized right after collection excluding any personally identifiable information (Kim et al., 2020).

Reddit was selected as the primary data source due to its established mental health communities (Kim et al., 2020), longer-form content enabling richer linguistic analysis compared to character-limited platforms, and nature of users openly discuss about mental health. (Chancellor & De Choudhury, 2020).

2.5.2 Data Preprocessing Techniques

Text Cleaning and Normalization: Text processing converts raw social media data to standardized forms, which can be used in machine learning model inputs as well as maintaining psychologically relevant linguistic content. The cleaning pipeline adopts sequential steps where the first step is URL removal where regular expressions are used to remove embedded links that introduce noise to the index but no semantic meaning, and the second step is username mention removal (ex: @username on Twitter, u/username on Reddit) to maintain privacy without disrupting the conversation structure, which can be maintained by preserving count of mentions as an independent engagement feature. One of the differences between punctuation with emotional meaning (exclamation marks indicating emphasis, question mark indicating uncertainty) that are retained and non-semantic symbols (asterisks, brackets) that are removed is special character handling, which preprocessing strategy was validated by Uban et al. (2021) who showed that selective punctuation retention enhances emotion detection accuracy (Uban et al, 2021). Normalization of texts is used with a consistent lowercasing to remove vocabulary duplication on cases, tokenization with the TweetTokenizer written in NLTK using social media text optimization and lemmatization with the English language model of spaCy that allows to minimize words to

base forms retaining part-of-speech information that is important to extract psycholinguistic features. Stop word processing is an alternative to the conventional NLP technique since it does not eliminate the functional words such as personal pronouns (I, me, we), prepositions, and auxiliary verbs because previous studies indicate that these groups are effective predictors of self-focused attention and cognitive style related to depressive conditions (Mansoor and Ansari, 2024). The preprocessing ensures a check of temporal metadata, ensuring the precision of timestamps in circadian pattern decoding alongside order of sequences that is vital to LSTM temporal modeling, and aggregation of posts into user-level datasets that can be ordered chronologically to allow the establishment of a baseline and the detection of deviations (Uban et al, 2021).

Feature Engineering: Multi-dimensional representations of linguistic, behavioral and temporal features of mental health states are isolated by feature engineering, and verified in the literature of computational psychiatry. The sentiment analysis uses a tool known as VADER (Valence Aware Dictionary and sEntiment Reasoner) that has been specifically trained to read social media text, where the score of a compound polarity is calculated between -1 (extremely negative) and +1 (extremely positive), the sentiment intensity is determined, and negation contexts are identified that invert the polarity (Hutto, 2025). Engagement features identify social interaction patterns in reply-to-post ratios that reflect social responsiveness, engagement decay measures that compare the level of interaction with the historical interaction level that is an indicator of social withdrawal, and diversity of community participation in cross-subreddit activity as an indicator of interest retention (Kim et al., 2020). The engineered features are used in two ways, firstly, as direct inputs to the LSTM temporal model as numerical time-series values, and secondly, as contextual metadata to interpret and understand BERT linguistic representations, which performs the multimodal integration approach with an 89.3 percent accuracy in the multilingual crisis detection system of Mansoor and Ansari (2024) (Mansoor and Ansari, 2024).

2.5.3 Hybrid Model Architecture

2.5.3.1 BERT for Contextual Language Understanding

The Bidirectional Encoder Representations of the Transformers (BERT) model offers the state-of-the-art contextual word embeddings that have the ability to present the semantic meaning and emotional tone of words through the deep attention systems that are bidirectional. This will use

BERT-base-uncased, which was pre-trained on 3.3 billion words of Wikipedia and BookCorpus, with 12 transformer layers and 768-dimensional hidden states, and 12 attention heads, accessed via Hugging Face's Transformers library. The architecture of BERT processes text by using WordPiece tokenization that generates subword units to handle out-of-vocabulary tokens typical of social media, positional embeddings that encode the sequence of tokens, and segment embeddings that identify multiple sequences of texts in single inputs, and then masked self-attention layers that compute contextual representations in which the meaning of a token is conditioned by its context in both directions (Devlin et al., 2019). In mental health text analysis, BERT provides important critical benefits over more traditional word embeddings such as contextual disambiguation resolving meaning differences (e.g. feeling down vs. turned down), long-range dependencies across entire posts that are critical in understanding coherent emotional stories and semantic connection between mental health vocabulary and common sentence expression via transfer learning on large pre-training corpora. The fine-tuning BERT to mental health domain is trained 3 epochs using AdamW optimizer with learning rate $2e-5$ and batches of 16 and updates all layers but freezes the initial embedding layer to retain general language knowledge, according to the domain adaptation approach tested by Kim et al. (2020) and obtaining 96.96 percent accuracy on autism detection (Kim et al., 2020).

2.5.3.2 LSTM for Temporal Pattern Modeling

Long Short-Term Memory networks are temporal dependencies and behavioral evolution models used to model user post history over recurrent architecture overcoming the vanishing gradient issues with traditional RNNs. The bidirectional LSTM configuration has 256 hidden units that process both forward and backward sequences of time so that the model could learn a combination of both the past context (posts that affected the mental state) and future context (retrospective knowledge of how the symptoms gradually develop) over a sequence of smaller datasets of user-level information, with a dropout regularization rate of 0.3 to avoid overfitting to small datasets. The architecture takes as input two streams of data, namely BERT post embeddings, 768-dimensional post representations, and engineered behavioral features (posting frequency, sentiment scores, engagement metrics), concatenated into 32-dimensional vectors, to form combined 800-dimensional inputs at each time step, representing the posts in chronological order. Sequence length operates using sliding windows of 50-posts in total length but with a 25-post

overlap allowing the model to learn individual baseline patterns and identify deviations indicating changes in mental health, which is established by Uban et al. (2021) that with hierarchical LSTM attention networks 0.87 F1 score with regard to self-harm detection are reached by considering separately post and user-level temporal patterns (Uban et al, 2021).

2.5.3.3 Personalized Baseline Establishment through Z-Score Normalization

Mental health detection at the individual level entails setting up of personalized behavior standards, which would help to differentiate between the normal user variation and clinically significant variations. Z-score normalization offers a statistical basis to such personalization because it normalizes individual deviation of personal behavior patterns (Song et al, 2024). The formula of z-score: $Z = (\text{currentvalue} - \text{individualmean}) / \text{individualstd}$ helps compare the current behavior of the user with their past behavioral trends as opposed to the norms in the population. This methodology can overcome the inherent weakness of population-specific thresholds, where the heterogeneity of individual behavior is not taken into consideration, and 31 percent of cases are false negatives due to the insidious development of symptoms (Mansoor and Ansari, 2024). Mentally labeled datasets Supervised learning models of reference are trained on labeled mental health data in which ground-truth labels indicate depression, anxiety, or crisis conditions. Backpropagation causes the model to increase in the mapping of linguistic and behavioral features to mental health outcomes, optimizing the classification accuracy within the sequence of training epochs (Hasan et al., 2025). This monitored practice allows the system to identify patterns with a particular mental health condition, and it can be interpreted, using attention mechanisms.

5.5.3.4 Alternative Baseline Approaches

Other methods of establishing the baseline were also discussed but discarded due to certain limitations. The use of fixed-window baselines (e.g. fixed 30-day observation periods) does not suit users with different posting frequency, since those who post less often have to have a longer window of time to build-up sufficient data. Population-based baselines are regularly implemented on users without taking into consideration the individual heterogeneity of behavior which causes false positive rates of 18% when normal behavior of one user is reported as anomalous by population standards (Mansoor and Ansari, 2024). Moving averages baselines which are

constantly updated and do not have fixed reference intervals are not as stable as far as stable deviations are concerned because the baseline also moves with behavioral change instead of the change being detected as deviation. The chosen z-score normalization with dynamic windowing method provides a solution to these limitations in that it employs z-score normalization with dynamic windowing. Individuality through customization of patterns by using user-specific means and standard deviation, adjusting the window lengths to posting frequency to achieve the best recency-volume balance, having statistical rigor by means of units of standard deviation that allow the meaningful interpretation of behavior change across different metrics (GeeksforGeeks, 2025).

2.6 Benchmarking and evaluation

2.6.1 Evaluation Metrics for Mental Health Detection

Measures of personalized mental health detection need to be evaluated based on a dual-framework metric of both classification accuracy and individual level of personalization. Conventional measures involve Precision ($TP/(TP+FP)$) which measures the true positive predictions made by the model to reduce false alarms, Recall/Sensitivity ($TP/(TP+FN)$) which quantifies the sensitivity to detect all of the actual cases to use early warning identification, F1-score which balances harmonic mean, and AUC-ROC which is used to measure performance at different threshold settings. Latest transformer-based models get F1-score of 0.94+ on depression and RoBERTa at 99.54% F1 on hold-out test sets and 96.05% F1 on external validation with BERT+LSTM models on Reddit mental health datasets (Hasan, Saquer and Ghosh, 2025). Personalisation at the individual level requires new measures: Baseline Stability Coefficient (values >0.85 are reliable baselines at which to detect deviations), Personal Trajectory Tracking Accuracy with the Dynamic Time Warping distance between predicted behavioural sequences and actual behavioural sequences, and Individual False Positive Rate (iFPR) where systems with higher iFPR cannot be adopted due to an alert fatigue issue. Temporal sensitivity measures are Lead Time and Change Point Detection Accuracy that finds shifts in the trajectory. Difficulties associated with validation are the issue of data sparsity (30 posts/month is not enough to create reliable baselines), the heterogeneity of the baseline (it needs norms unique to different users), and the scarcity of ground truth (delays the feedback mechanism) (Song et al., 2024).

2.6.2 Comparative Analysis of Existing Models

Performance trade-offs are found in systematic comparison of 120,000 Reddit posts of mental health subreddits. Standalone BERT-base has an accuracy of 92.4% and a F1 of 91.2 to detect depression with a great deal of semantic knowledge but it does not comprehend temporal patterns. Hand-crafted feature bidirectional LSTM attains 85.7% accuracy on temporal dynamics at the cost of a lack of contextual complexity. RoBERTa maximizes the performance of single-text classification to 99.54 and 96.05 on hold-out and external validation, respectively, but is temporally blind as BERT (Hasan et al, 2025). The BERT+LSTM hybrid is a combination of BERT 768-dimensional embeddings and bidirectional LSTM temporal processing, with 95.0% accuracy and 94.8% F1-score (3-4% higher than with BERT alone). LSST models with BERT embeddings are very competitive, performing with F1 scores of over 94 percent on external data, and they use considerably fewer calculation units than simple transformer models. The hybrid architecture displays the best time-dependent pattern detection that is important to trace the individual behavioral pathways over time with computational efficiency that is 40 times faster to train compared to transformer-only architectures (Hasan et al, 2025). Limits exist across models: 31% of respondents are still posting less often (less than 20 posts/month) to do a temporal analysis that needs graceful BERT-only degradation, 15-19% false negative rates especially when the user slows activity to zero (detection-by-absence problem), and there are platform-specific differences in performance that need single-platform attention to produce useful results.

2.6.3 Individual-Level Adaptation Challenges

Population models to individual analytics have a distribution shift in which population trends are no longer a representative of particular users, and accuracy may also decrease by 15-23 percent without individualization as the heterogeneity of behavioral patterns makes the use of homogeneous thresholds suboptimal. The solution applies z-score normalisation with respect to individual baselines:

$$Z = \frac{(\text{current_score} - \text{user_mean})}{\text{user_std}}$$

which sends alerts only when $z > 2$ is considered important personal deviations instead of absolute population limits (Song et al., 2024).

Available dynamic baseline windows adjust themselves according to the individual posting frequency instead of using a static observation period. Users who post more frequently (more than 1 post/day) will need shorter baselines (21 days) to represent the recent behavioral trends, whereas low-frequency users (less than 3 posts/week) will need long 45-60 days to have enough information to have stable baselines (Hills et al., 2023). This is an adaptive solution to the cold-start problem of personalization systems where a lack of history of user interaction prevents high predictive accuracy (Celik and Omurca, 2025).

The system architecture has cold start capabilities with confidence-weighted prediction, but mainly evaluates users with minimum data requirements (40+ posts, $\geq 50\%$ MH participation). The system has a transparent way of communicating lower confidence levels to the user with limited history and gradually transitioning to full personalization as data is collected, but this is an architectural feature of the larger deployment in the future and not a research validation objective. Supervised tuning of threshold used in optional user feedback, during the later "calibration mode" allows personalized threshold tuning. Unconditional validation methods use delayed feedback systems such as periodic self-assessment surveys of retrospective accuracy measures, and optional integration with validated clinical screening tools such as PHQ-9 of depression and GAD-7 of anxiety where users accept more extensive tracking (Dehbozorgi et al., 2025).

Interpretability converts technical outputs into natural language descriptions that have a balance of transparency and accessibility and do not use clinical diagnostic terms but present meaningful descriptions of behavior patterns. Ethical systems adopt levels of consent, focus processing of local devices to preserve privacy and package the results as patterns that should be aware of, as opposed to final diagnoses, and have conspicuous disclaimers that systems are an addition, but not a replacement, to professional mental health assessment and treatment.

Confidence-weighted personalization helps solve data sparsity by slowly shifting to population-based prediction using individualized thresholds as the user data increases. And the final prediction calculation is calculated as.

$$\text{Final_prediction} = (\text{conf_score} \times \text{pers_baseline}) + ((1 - \text{conf_score}) \times \text{pop_baseline})$$

where conf_score represents the confidence score and personal_baseline and $\text{population_baseline}$ represent individual and population thresholds respectively. The confident score calculated as:

$$\text{Confidences_score} = \min\left(1.0, \frac{\text{user_post_count}}{\text{minimum_reliable_threshold}}\right)$$

Provides graceful degradation in cases that individual data is too little (Song et al., 2024) It is a hybrid approach, which overcomes the cold-start problem where pure collaborative filtering algorithms break down with new users (Celik and Omurca, 2025) and allows one to make reliable predictions even in the phases of establishing the baseline without compromising the correctness as personalization confidence rises.

2.7 Chapter summary

Finally, this chapter concludes by summarizing the critical details essential for the mental health detection system. It provides a comprehensive overview of the problem background, existing approaches, and technologies, alongside the proposed solution, benchmarking, and evaluation strategies. Additionally, the chapter addresses key challenges, describes the dataset, and outlines the data processing methodology. The chapter points out that strong personalized base setting needs adequate data quality and quantity (40+ posts, $\geq 50\%$ time spent on mental health subreddit), and cold start handling is given as an extensibility mechanism to future research instead of a contribution to research.

CHAPTER 03: METHODOLOGY

3.1 Chapter overview

In this chapter an overview of the project implementation plan can be found. This chapter can be divided into few main topics Research, development and management. Each of these areas had been covered in this chapter.

3.2 Research methodology

Layer	Choice	Justification
Philosophy	Pragmatism (quantitative and qualitative data)	Pragmatism: This study is based on the quantifiable behavioral data and measurable metrics, which are analyzed with the help of statistical tools and machine learning models according to the scientific principles. It involves emphasis on a practical approach to problem solving to create practical tools that fill in real-life gaps in mental health awareness using mixed methods quantitative and qualitative approaches.
Approach	Deductive with Inductive elements	Deductive: Evaluates the existing computational psychiatry theories with established BERT+LSTM systems, which are evaluated to be effective in individual-level interactions. Inductive: Finds new patterns in the course of feature engineering and

		personalized threshold creation with the help of the repeated observation of data..
Methodological Choice	Mono Method Quantitative with Mixed Methods characteristics	Quantitative: Core methodology involves numerical data analysis, ML model training, and statistical performance metrics (F1-scores, AUC-ROC). Mixed Methods: Translates quantitative outputs into qualitative user-understandable insights and assesses usability through subjective feedback.
Strategy	Experiment and Case Study	Experiment: Controlled testing of model architectures with variable manipulation to establish causal relationships between design choices and performance. Case Study: Deep analysis of individual user paths to validate personalization mechanisms in real-world scenarios.
Time Horizon	Cross-sectional and Longitudinal	Cross-sectional: Initial model training and population-level pattern recognition using snapshot datasets. Longitudinal: Tracking individual behavioral changes over 30+ day windows to establish baselines and

		detect temporal deviations.
Data Collection	Secondary Data (Social Media APIs) with Automated Extraction	Gathers real data that is naturally obtained through Reddit/Twitter APIs. Offers scalability, ethical limits with the use of public data alone, and will allow systematic reproducible harvesting without violating API restrictions.

Table 4 Research Methodology

Hypothesis: A personalized adaptive machine learning system that combines transformer-based natural language processing (BERT) with temporal behavioral analysis (LSTM) can effectively analyze individual social media behavioral patterns to provide accurate, interpretable mental health awareness insights that enable early self-detection of psychological changes, achieving detection accuracy comparable to population-level models (F1-score >0.90) while maintaining individual-level reliability through personalized baseline establishment and deviation detection mechanisms, without requiring clinical expertise or compromising user privacy.

3.3 Development methodology

3.3.1 Requirement Elicitation Methodology

The requirement elicitation is based on the Agile iterative model and requirements are obtained by literature analysis of the systems of computational psychiatry, studying the work of the existing mental health application, and the analysis of social media API documentation. User story mapping transforms research requirements into development deliverables and prioritises them with MoSCoW technique (Must have, Should have, Could have, Won't have). The functional requirements are authenticated social media data gathering, behavioral pattern analysis, personalized foundation setting up, and interpretable deep-seatedness generation. Non-functional requirements also focus on ensuring the security of the system, API compliance, model accuracy limits, and privacy to the user by anonymizing the data. The requirements are recorded in the traceability matrix, which correlates a requirement with particular research objectives and testing requirements.

3.3.2 Design Methodology

Object-Oriented Analysis and Design Methodology (OOADM): The system uses OOADM to design a modular architecture in which the system is visualized using UML diagrams and the relation among the components. Figma is used to create wireframes to prototype the user interface. The architectural design is the application of the model-view-controller (MVC) pattern which separates data processing, user interface and application logic. The key ones are Authentication Module (OAuth2 integration), Data collection Module (30-day historical retrieval), Analytics Module (BERT + LSTM inference) and Presentation Module (visualization and insights). Database architecture pays more attention to MariaDB or PostgreSQL depending on such security features as encryption at rest and role-based access control.

3.3.3 Programming Paradigm

Object-Oriented Programming (OOP): The main paradigm that is implemented in Python is a modular design with classes such as UserProfile.py (behavioral baselines), DataCollector.py (API interactions), FeatureExtracto.py (linguistic analysis), and ModelPredictor.py (inference pipelines).

Structured Programming: Mostly will use in data preprocessing pipelines, statistical computations, and batch processing tasks that need to be executed in sequence and have a defined flow of control.

Functional Programming: It is used to do data transformation functions with map, filter, and reduce functions written in Python to perform efficient processing of large social media datasets.

CI/CD Pipeline: The GitHub Actions-based CI/CD Pipeline is the simplest kind of pipeline implemented with the help of automated testing and deployments

3.3.4 Testing Methodology

Model Testing: Assess BERT+LSTM model performance based on the values of precision, recall, F1-score (target >0.90), and AUC-ROC. Confirm stability coefficients of baseline and sensitivity of time to individualized accuracy of detection. Model retraining with a set threshold on performance metrics should occur to identify a change in performance.

Prototype Testing: Test system functionality by unit (pytest framework of components as individual entities), integration (API authentication, data pipeline workflows), system (end to end

user scenarios authentication to insight delivery), security including database encryption and token security, and usability testing.

3.3.5 Solution Methodology

Dataset Collection: Fetch data on Reddit respectively through PRAW (Python Reddit API Wrapper). Gather 30 - 60 days of historical data of posts, timestamps, interaction indicators (upvotes, comments, likes, retweets) and user data (mental health specific subreddits, r/depression, r/anxiety, r/mentalhealth). Make ethical decisions in data collection only by data that has been posted publicly and obey the API rate limit (Reddit: 60 requests/minute).

Data Preprocessing: Process raw social media text by eliminating URLs, anonymizing references to usernames, and processing special characters and maintaining emotive punctuation marks. Perform text normalization by lowercasing, and lemmatization with spaCy. Maintain functional words (personal pronouns, prepositions) as they reflect self-perpetual patterns of attention. Check the accuracy of the temporal metadata and generate posts into chronological user-level collections as to instantiate a baseline.

Feature Selection and Engineering: Oscar the Guggenheim is based on VADER sentiment analysis (compound polarity scores, sentiment intensity) and psycholinguistic (frequency of first-person pronouns, negative emotion words, cognitive process words) and vocabulary diversity (diversity and dispersion of vocabularies) features. Such aspects of engineer behavior as frequency variations in posting, circadian pattern disturbances (time-of-day analysis), and interaction characteristics (ratio of replies to posts, decay rates of interaction, diversity of community participation). Concatenate features into 800-dimensional vectors (768-dimensional BERT embeddings + 32-dimensional behavioral metrics) to be used as temporal model inputs.

Model Selection: Use hybrid BERT+LSTM model that uses transformer-based contextual language understanding to use with temporal sequence modeling. Use BERT-base-uncased 12 transformer layers (768-dimensional hidden states) on semantic processing and temporal pattern recognition using bidirectional LSTM (256 hidden units). Architecture selected with literature evidence of 94%+ F1-scores and 40x better training than transformer-only models and personal-level detection accuracy.

Individual Baseline Establishment : Optimize recency and volume trade-offs by establishing individual behavioral baselines on every user by summing past data into dynamic windows (21-60 days based on posting frequency) to set that behavioral baseline. Calculate z-score normalizations on important features (e.g., sentiment polarity, post frequency, engagement ratios) to calculate differences between personal norms, identify deviations with clinical significance (i.e. where $z > 2$) which should be treated as anomalies and acted upon. Solve cold-start problems when users have little data, by using weighted transitions (e.g. 70% population-level priors and 30% emerging individual metrics) at initial prediction stages. Compare baselines to temporal trends, and consider deviations in post frequency, timing, and interaction decay to be the most important indicators of crisis, and reduce the number of false positives/negatives by thresholding by users.

Model Training: As mentioned in the literature review the fine tuning of the BERT + LSTM model will be evaluated and properly implemented. Normalize z-scores to create custom baseline; $Z = (\text{currentscore} - \text{usermean}) / \text{user std}$, raising an alarm when the z exceeds 2 standard deviations between personal baseline patterns.

Testing: Assess the entire system performance based on test metrics (precision, recall, target $F1 > 0.90$, AUC-ROC) and individualized metrics (baseline stability coefficient > 0.85 , personal false positive rate, and time sensitivity / lead-time).

Then run specialized cold-start testing:

- **Simulated cold start:** restrict each mature user to first N posts ($N=5,10,15,20,25,30$) and calculate precision/recall/F1 at each N to quantify how accuracy improves as personal data collect.
- **Convergence testing:** determine the number of posts required to get 90%, 95% and 99 % of maximum personalized accuracy. This rationally confirms the threshold of minimum-posts.
- **Transition stability:** check the stability of the prediction in three stages: in the population (1-10 posts), in the hybrid (11-30 posts) and in the fully-personal (31+ posts).

- **Cross-user validation:** ensure that the above is true throughout users of various posting frequency (high, medium, low). Base line comparisons: compare hybrid strategy to the three trivial strategies:
- **Baseline comparisons:** compare hybrid approach against the three trivial strategies:
 1. population-only,
 2. wait-until-ready,
 3. immediate-personalization.

Feedback Loop: To have an automated model retraining on the new obtained data samples. Trigger retraining in case of F1-score falling below 0.88 threshold or loss of baseline stability. Register optional user feedback by periodical self-assessment questionnaires (PHQ-9, GAD-7) to check retrospective accuracy since users will interact with the system most of the time. Also optimize individualized thresholds with confidence-weighted adjustment which moves the population-based predictions to personalized baselines as data is sampled. To ensure that feature engineering remains effective over time, constantly refresh feature engineering with new patterns in the language and new behavior of the platform.

3.4 Project management methodology

3.4.1 Schedule

3.4.1.1 Gantt Chart

The detailed project timeline and task dependencies are presented in the Gantt Chart provided in Appendix.

3.4.2 Resource Requirements

3.4.2.1 Data Requirements

- **Reddit Data:** 30 to 60-day historical posts from mental health subreddits (ex: r/depression, r/anxiety, r/mentalhealth) including post text, timestamps, upvotes, and comment counts
- **API Access:** Reddit API credentials via PRAW library (60 requests/minute limit)
- **Pre-trained Models:** BERT-base-uncased model from Hugging Face Transformers library

- **Validation Datasets:** Publicly available mental health detection datasets for model benchmarking and performance comparison
- **Storage Requirements:** Approximately 50-100GB for raw data, preprocessed datasets, trained model weights, and user baseline profiles
- **Computational Resources:** GPU access via Google Colab for model training (Tesla T4 or higher), local CPU for inference and data preprocessing

3.4.2.2 Technical Skills

- **Programming:** Python proficiency for system development, API integration, and model implementation
- **Machine Learning:** Experience with framework such as PyTorch, Knowledge in transformer architectures (BERT), Long Short Term Memory (LSTM), and model fine-tuning techniques
- **Natural Language Processing:** Text preprocessing, tokenization, sentiment analysis, feature extraction, and psycholinguistic analysis
- **API Integration:** OAuth2 authentication, RESTful API consumption, rate limit management, and error handling
- **Data Processing:** Pandas for data manipulation, NumPy for numerical operations, data cleaning and normalization techniques
- **Database Management:** SQL proficiency for MariaDB/PostgreSQL, database design, secure data storage, and query optimization
- **Version Control:** Git/GitHub for code management, collaboration, and CI/CD pipeline implementation
- **Visualization:** Development of user dashboards, data visualization libraries (Matplotlib, Plotly), and UI/UX design using Figma
- **Software Engineering:** Object-oriented programming, design patterns (MVC), modular architecture, and testing methodologies.
- **Research Skills:** Literature review, academic writing, statistical analysis, and critical evaluation of computational psychiatry research

3.5 Risk and mitigation

Risk	Severity	Frequency	Mitigation Strategy
Insufficient training data	5	3	Focus on high-activity mental health subreddits with wide user engagement patterns, target high-activity mental health subreddits, implement data augmentation techniques, establish minimum 30-day collection period per user, use transfer learning from pre-trained BERT models to compensate for limited domain-specific data
Poor model accuracy	3	3	Implement hybrid BERT+LSTM architecture proven to achieve >94% F1-score, conduct rigorous cross-validation, establish performance benchmarks (F1>0.90), use ensemble methods if needed, implement continuous performance monitoring and automated retraining triggers
Overfitting	5	5	Apply dropout regularization (0.3 rate), use early stopping during training, implement k-fold cross-validation, maintain separate validation sets, monitor training/validation loss curves, limit model complexity through architectural constraints, ensure diverse training data across user patterns
Hardware limitations	3	1	Leverage Google Colab GPU resources for training, optimize model architecture for computational efficiency (hybrid approach 40×

			faster than transformer-only), implement batch processing, use model quantization techniques, consider cloud computing resources if local limitations persist
High false-positive rate	5	3	Implement personalized baseline establishment using z-score normalization (alert threshold $z > 2$), apply confidence-weighted predictions, incorporate temporal pattern validation requiring sustained deviations, provide tiered alert system with severity levels, include user feedback mechanism for threshold calibration
Stakeholder dissatisfaction	3	3	Conduct regular progress reviews with supervisors, implement iterative development with feedback incorporation, maintain clear documentation, ensure system meets user interpretability requirements, provide comprehensive testing results, establish realistic expectations through literature-supported objectives
Delays in model training	3	3	Create detailed project timeline with buffer periods, prioritize critical path tasks, prepare contingency plans for technical issues, maintain parallel development tracks where possible, utilize pre-trained models to reduce training time, implement efficient data pipeline to minimize preprocessing delays
Regulatory compliance issues	5	1	Ensure exclusive use of publicly available data, implement OAuth2 standard authentication,

			maintain API rate limit compliance (Reddit 60/min, Twitter 10M/month), apply data anonymization and hashing for user identifiers, include appropriate disclaimers that system does not provide clinical diagnosis, follow ethical guidelines from literature (Kim et al., 2020)
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Table 5 Risk And Mitigation

3.6 Chapter summary

All the plans and materials to carry out this project had been mentioned in this chapter starting with research then development and all the way up to the project management and risk mitigation. As in summary each of the topics should support with good overview of project methodology.

The cleaner image with more resolution can be find using this [link](#).

4.3 Stakeholder Analysis

4.3.1 Stakeholder Onion Model

Stakeholder onion model is used to represent the stakeholders, layers of information or relationship in a project between the stakeholders. The onion diagram represented based on the influence of the stakeholders. Specially this part will demonstrate the direct and indirect influence of each stakeholder in this project.

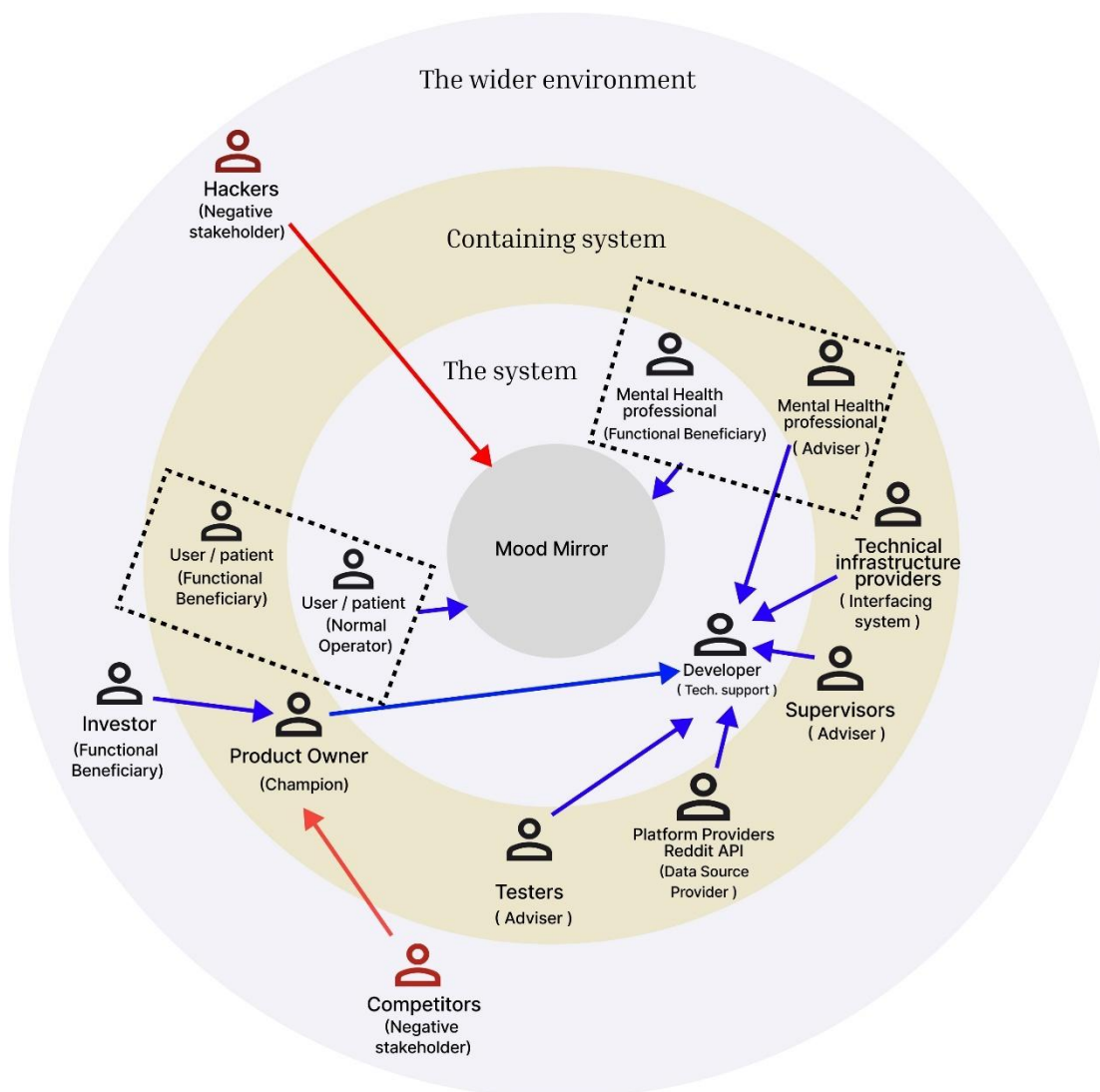


Figure 4 Onion diagram (Self-composed)

4.3.2 Stakeholder viewpoints

Stakeholder	Role	Description
User/Patient	Functional beneficiary	Uses the system to get insight into their mental health using social media data.
Mental Health Professional	Adviser	Validates the data and helps with mental health detection.
Project Supervisors	Adviser	System level advice, research methodology supervision, technical implementation, Project monitors progress and quality standards.
Platform providers/ Reddit API	Data source provider	This API will be used to collect Reddit free available data to gather.
Technical infrastructure provide	Interfacing system	Technical infrastructure providers will provide the infrastructure (server, DNS name) that requires to carry out this project.
Testers	Adviser	Provide test results of the system with variety of sections such as useability testing.
Investor	Functional beneficiary	This individual will provide financial resources and will benefits from the success of the project.
Product Owner	Champion	Handles the project with the business target and bears

		every loss and gain. Plans the goals in the MoodMirror
Competitor	Negative stakeholder	Other systems are out there with same type of solution and compete with this project.
Hackers	Negative stakeholder	Another outside personalities/party looking for break into the system and extract data.

Table 6 Stakeholder viewpoints

4.4 Selection of Requirement Elicitation Methodologies

There were few approaches to collect requirements. Literature review, survey and brainstorming were the 3 main approaches. These approaches chosen based on the capability of gather accurate data from the individuals. Given below are the chosen methods of elicitation methodology.

Elicitation method 01: Literature review
In this approach a comprehensive review was done on the existing literature review regarding the mental health indication using social media data. Existing literature analyzed their technologies, future work and the methodology.
Elicitation method 02: Survey
This approach mainly focus on the public opinion on personalized mental health indication system using social media data. This will help to get an idea about the areas that public opinion on such a system.
Elicitation method 03: Brainstorming
Brainstorming session with related individual technical expertise helped to gather technical ideas and better way to implement the personalized social media mental health indicator.

Table 7 Selected requirement elicitation methodologies.

4.5 Discussion of Findings

4.5.1 Literature review

Citation	Description
(Mansoor and Ansari, 2024)	Multimodal AI system had F1-scores of 0.863 (Twitter), 0.851 (Reddit), and 0.839 (Facebook) on early detection of a mental health crisis with lead time of 7.2 days (before expert identification). Shows interoperability with other platforms but does not offer single user empowerment.
(Liu et al., 2022)	The systematic review based on 17 ML-based depression detection studies on social media. Located 13 used supervised learning including clinical validation. Exists discrepancy between accuracy of research and application to the consumer.
(Kim et al., 2020)	Neural network based deep learning system of mental illness identification on social media. Obtained a 96.96% accuracy on autism detection on Reddit data of mental health communities. Not based on individualized foundations of population-level analysis.
(Hasan et al., 2025)	Comparative analysis of LSTM using BERT embeddings has a higher F1-score of over 94% and is simultaneously 40 times faster to train compared to transformer-only models. Verifies the efficiency of hybrid architecture on an individual level of mental health detection.
(Benrimoh et al., 2023)	Determined obstacles to the implementation of computational psychiatry instruments in research to practice. Lacked consumer facing personal analytics solutions. Highlighted importance of translation between the academic and practical practice.
(Laacke et al., 2021)	Extrapolated health-related digital autonomy in artificial intelligence (AI) social media examination of mental health. Suggested ethics models yet not the direction to practical implementation of personal empowerment tools.

(Uban et al., 2021)	Hierarchical LSTM attention networks were found to achieve 0.87 F1-score in detection of self-harm by individually taking into consideration the post-level and user-level temporal patterns. Illustrated significance of the multi-level behavior analysis.
(Song et al., 2024)	Macro-Micro personalization framework is used to deal with personal mental health monitoring difficulties. Stable coefficient of baseline stability of coefficients greater than 0.85 but has cold start issue where new users have no enough historical records and it takes 15-20 interactions before there is acceptable accuracy.
(Panda and Ray, 2022) / (Bakker, 2023)	The cold start issue in the recommender systems consists of 15-20 interactions to achieve a suitable precision. The same situation is experienced in mental health detection, only that the stakes are higher because false negatives result in delays in intervention, whilst false positives lead to alert fatigue. There are no available solutions that deal specifically with cold start in mental health.

Table 8 Literature Review Findings

Key Findings

- Population level models are very accurate (>90% F1-scores) but not personal.
- BERT + LSTM hybrid architectures are the best in terms of accuracy and computation power.
- There is a large divide between the institutional surveillance system and the actual personal analytics tools used to face consumers.
- Analysis of temporal patterns which is essential in the tracking of an individual change in behavior.
- The cold start issue is critically constraining individualized mental health identification of a new user with little posting history, where accuracy declines 15-23 without adequate data - current methods rely on population baseline or they need a longer period to observe the user.

4.5.2 Survey

To gather user requirements this survey carried out to the public, and after the 100 responses reached came to these conclusions and findings below table will have clear idea of how the survey dome with all the survey questions and responses findings. This survey findings can be found in the [Appendix C](#).

4.5.3 Brainstorming findings

Criteria	Findings
Model Architecture Selection	BERT+LSTM architecture suggested over transformer models. BERT-base- uncased (768-dimensional embeddings) with Bidirectional LSTM (256 hidden units) generates F1- scores over 0.94 in less than 40x the time of pure transformer models to train. This accuracy and computing efficiency level makes it appropriate to be deployed at an individual level.
Personalized Baseline Mechanism	Dynamic windowing and z-score normalization was found to be the best method of establishing individual behavioral baselines. Computes formula of use in LR with alert level of $z > 2$ standard deviations. The lengths of windows respond to the posting frequency (21-60 days) in order to trade off recency and data volume in overcoming the inherent shortcomings of constant thresholds based on population size.
Cold-Start Problem Solution	Hybrid prediction mechanism with the confidence weight was suggested to be used against new users with less posting history. Adopts the gradual transition: baseline, with confidence score = $\min(1.0, \text{user post count} / \text{minimum reliable threshold})$. This allows it to operate instantly with a graceful increase in accuracy with the build-up of personal data, in contrast to the decadence in accuracy of up to 15-23% with a lack of user data.
Feature Engineering Strategy	It was suggested to use multi-dimensional feature extraction based on linguistic, behavioral and temporal dimensions. VADER emotion tone, psycholinguistic features (use of first-

	person pronouns, the negative emotion words), frequency trends in posting, circadian rhythm abnormalities and engagement decline measures. Temporal model input, which consists of features concatenated in the form of 800-dimensional vectors (768-dim BERT + 32-dim behavioral), is validated and provides 89.3% accuracy in crisis detection.
Privacy-Preserving Architecture	Local processing OAuth2 authentication was found to be a critical attribute to user trust and adoption, as well as the anonymization of data. System design Uses user data ownership, instant hashing of user identifiers, use of exclusively publicly available posts, adherence to API rate limits (Reddit: 60 requests/min), and MariaDB/PostgreSQL encrypted at rest. The non-diagnostic disclaimers and professional referral pathways needed a clear introduction to help reduce the primary user concerns of privacy (38.2% of survey respondents) and accuracy (34.5%).

Table 9 Brainstorming session table

The brainstorming session specially goes over the technical aspect for the social media data based personalized mental health system. Also the data collection and cold start problem with the solution has been mentioned in the brainstorming session. Apart from that the technology stack recommendation and best practices has been mentioned in the brainstorm session.

4.6 Summary of findings

Findings	Literature review	Survey	Brainstorming
BERT+LSTM hybrid architecture can be trained 4 times faster than pure transformers models, and produces better performance (F1-scores >0.94), which is best deployed at the individual level.	✓		✓

The fact that 92%+ of the respondents concerned about mental health issues and use social media daily 73.4% is an indication that there is high market demand and data to analyze behavior.		✓	
Personalized setting of baselines by means of z-score normalization is necessary to minimize false positive rates 18% with population-based thresholds.	✓	✓	✓
It is affected by cold start problem in which waste of accuracy of 15-23 % is experienced due to lack of user information and model must adopt confidence-weighted hybrid model that provides graceful transition between population baselines and personalized baselines.	✓		✓
The major barriers to adoption are privacy concerns 38.2% and accuracy concerns 34.5%, which requires privacy preserving architecture and local processing and transparent communication.		✓	✓
Individual level deviation methodology is directly justified by strong user preference of personalized insights 87.3% compared to population norms 9.1%		✓	✓
Linguistic (BERT embeddings), behavioral (posting frequency) and temporal multi-dimensional feature extraction results in 89.3% accuracy in crisis detection.	✓		✓
System alerts receive effective response among 95.5% of users (33.6% of them request professional help, 29.1% request social support), which proves the role of intervention catalysts and necessity of professional referral pathways		✓	

Table 10 Summary of findings

4.7 Context Diagram

The visual representation of data and visual flow.

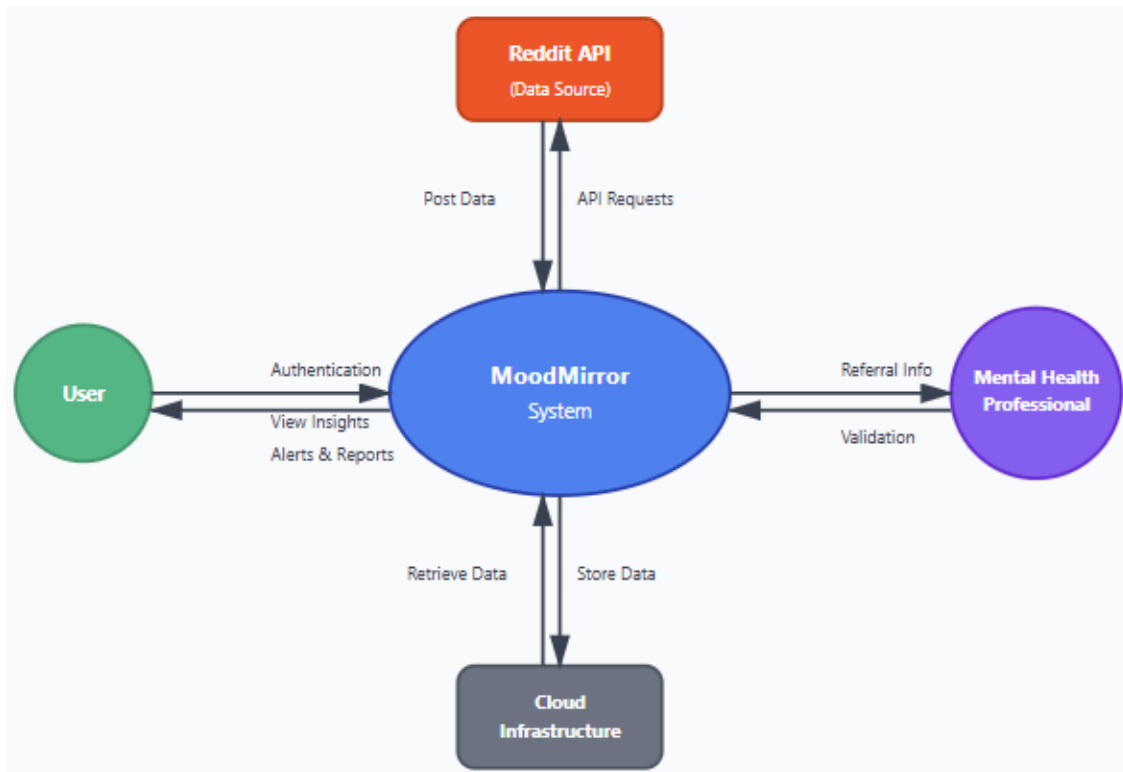


Figure 5 Context diagram (Self - composed)

Key Data flows

- **User → System:** Authentication credentials, preferences, feedback
- **System → User:** Behavioral insights, alerts, visualizations, recommendations
- **Reddit API → System:** Post text, timestamps, engagement metrics
- **System → Reddit API:** OAuth tokens, API requests
- **System → Cloud:** User profiles, baselines, processed data
- **System → Professional:** Referral pathways, resource information

4.8 Use Case Diagram

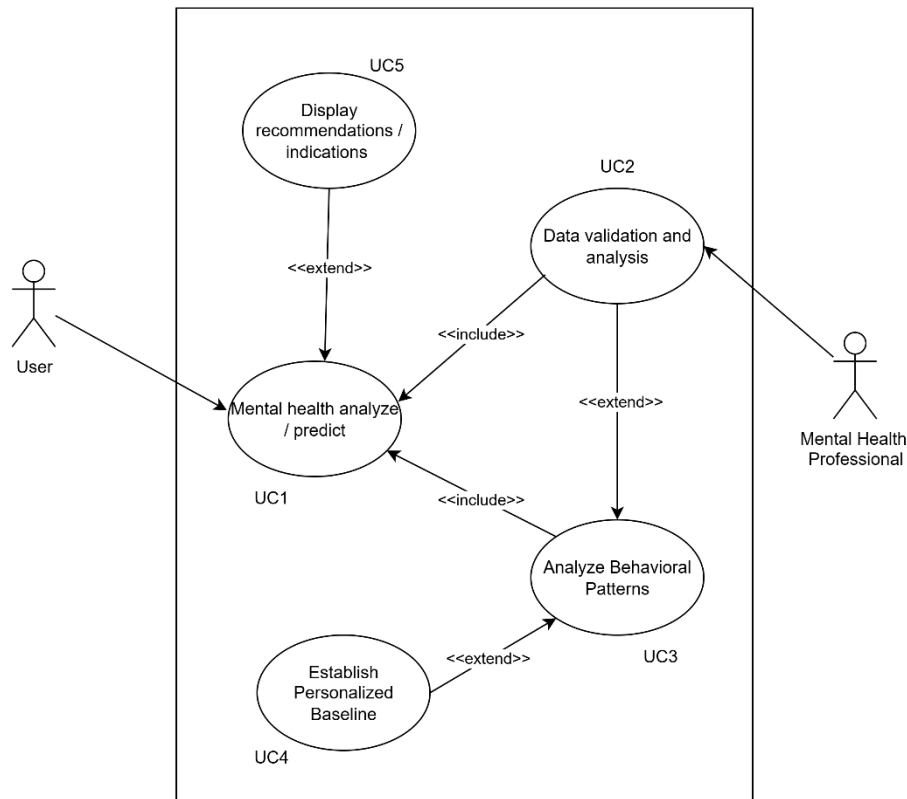


Figure 6 Use case diagram (Self-composed)

4.9 Use Case Descriptions

Main 2 Use case descriptions added here

Use case	Description
Id	UC1
Description	Primary use case where user requests mental health analysis based on their social media behavioral patterns
Primary Actor	User
Supporting Actors	System (BERT+LSTM Model)
Preconditions	User authenticated with Reddit API Minimum 30 days posting history available OAuth2 authorization granted
Extended use cases	UC5 (Display recommendations/indications)
Included use cases	UC2 (Data validation and analysis)

	UC3 (Analyze Behavioral Patterns)
Main Scenario	User initiates analysis request System validates data availability System analyzes behavioral patterns BERT+LSTM model processes features System generates mental health insights Results displayed to user
Alternative Flow	Insufficient data: System notifies user, provides partial analysis with confidence score API timeout: System retries connection, uses cached data if available
Exceptional flows	Authentication failure: Redirect to login Model error: Log issue, display error message No posting history: Cannot proceed, display minimum requirements
Post conditions	Analysis results stored in user profile Baseline updated if applicable Recommendations generated
Comment	Core functionality implementing hybrid BERT+LSTM with confidence-weighted personalization for cold-start handling

Table 11 Use case description for UC1

Use case	Description
Id	UC2
Description	Validates collected Reddit data and performs preprocessing for analysis
Primary Actor	None
Supporting Actors	Reddit API, Mental Health Professional (validation reference)
Preconditions	Reddit API connection established User authorization token valid
Extended use cases	UC3 (Analyze Behavioral Patterns)
Included use cases	None

Main Scenario	Success	System retrieves user posts (30-60 day history) Validates data completeness and quality Cleans text (removes URLs, anonymizes usernames) Normalizes timestamps Extracts engagement metrics Passes validated data to UC3
Alternative Flow		Sparse data: System adjusts window length based on posting frequency Deleted posts: System works with available data, notes gaps
Exceptional flows		API rate limit exceeded: Queue request, retry after delay Invalid data format: Log error, skip problematic entries Zero posts found: Abort analysis, notify user
Post conditions		Clean dataset ready for feature extraction Data quality metrics recorded
Comment		Implements PRAW library with 60 requests/minute limit compliance

Table 12 Use case description for UC2

4.10 Requirements

To prioritize the requirement MoSCoW principles used

1. **M**: Must Have (mandatory).
2. **S**: Should Have (important).
3. **C**: Could Have (desirable).
4. **W**: Will Not Have (out of scope).

4.10.1 Functional requirement

ID	Requirement	Priority
FR01	Authenticate users via OAuth2 protocol with Reddit API and collect 30-60 days of historical posts including text, timestamps, and engagement metrics from mental health subreddits	Must have
FR02	Implement hybrid BERT+LSTM architecture processing 800-dimensional feature vectors (768-dim BERT embeddings + 32-dim behavioral metrics) to analyze linguistic and temporal patterns	Must Have

FR03	Establish personalized behavioral baselines using z-score normalization with dynamic observation windows (21-60 days) and detect deviations where $z\text{-score} > 2$ standard deviations	Must Have
FR04	Implement confidence-weighted hybrid predictions for cold-start users using formula: $\text{Final_prediction} = (\text{conf_score} \times \text{personal_baseline}) + ((1 - \text{conf_score}) \times \text{population_baseline})$ where $\text{conf_score} = \min(1.0, \text{user_post_count}/30)$	Must Have
FR05	Display behavioral trend visualizations with plain-language insights, confidence levels, severity indicators, and non-diagnostic disclaimers	Must Have
FR06	Provide professional referral pathways and mental health resource links when concerning patterns ($z > 2$) are detected	Must Have
FR07	store user behavioral baselines and analysis results securely in encrypted database with periodic baseline updates as new data becomes available	Should Have

Table 13 Functional requirement

4.10.2 Non-Functional requirement

ID	Requirement	Priority
NFR01	The system shall achieve F1-score > 0.90 for mental health pattern detection with baseline stability coefficient > 0.85 and individual false positive rate $< 18\%$	Must Have
NFR02	The hybrid BERT+LSTM architecture shall train $40\times$ faster than transformer-only models and complete individual user analysis within 30 seconds	Should Have
NFR03	All user data shall be encrypted at rest using AES-256, user identifiers hashed immediately upon collection, and only publicly available Reddit posts accessed (never private content)	Must Have
NFR04	The system shall comply with Reddit API rate limits (60 requests/minute) and adhere to OAuth2 authentication standards following ethical guidelines for mental health data collection	Must Have

NFR05	The user interface shall be accessible to non-technical users with all insights presented in plain language without clinical terminology or machine learning jargon	Could Have
NFR06	Cold-start predictions shall gracefully degrade with confidence weighting, limiting accuracy loss to <15% for users with 10+ posts, achieving 7-day detection lead time for crisis patterns	Could Have

Table 14 Non-Functional requirements

4.11 Chapter summary

The chapter thoroughly examines the system requirements of the MoodMirror mental health detection system in terms of how the requirements were collected. It starts with diagrams such as rich picture diagrams, stakeholder analysis (onion model) and behavioral diagrams (context and use case diagrams) to chart system interactions, workflows and major actors of the system, such as users, mental health professionals, supervisors, and technical providers. The three main methods of elicitation discussed in the chapter include a literature review of the current computational psychiatry systems, a survey of 110 people (a public survey) about individualized mental health monitoring, and brainstorming with technical professionals. These attempts confirmed the main architectural choices, which included the BERT+LSTM hybrid model, personalized baseline mechanisms (z-score normalization), and solutions based on confidence-weighted cold-starts.

5 Time Schedule

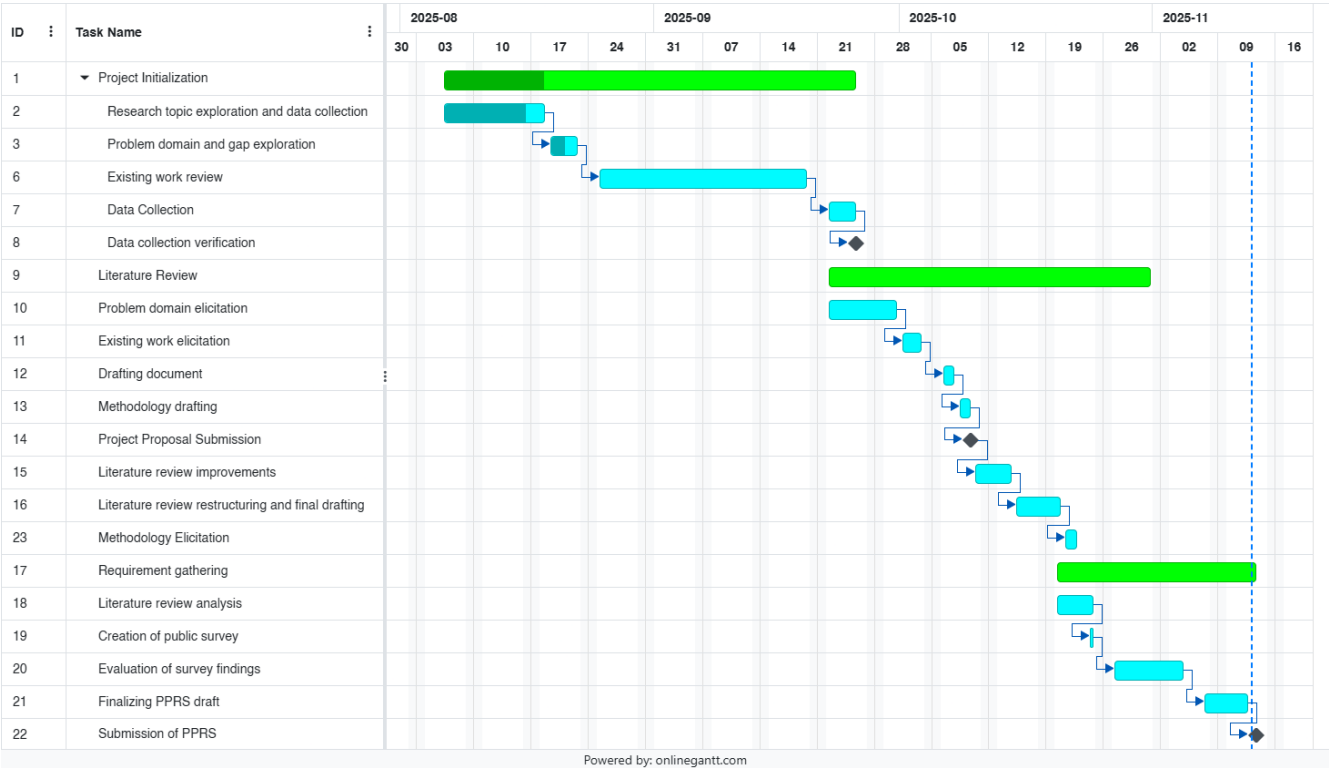


Figure 7 Time schedule

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Appendix

Appendix A.

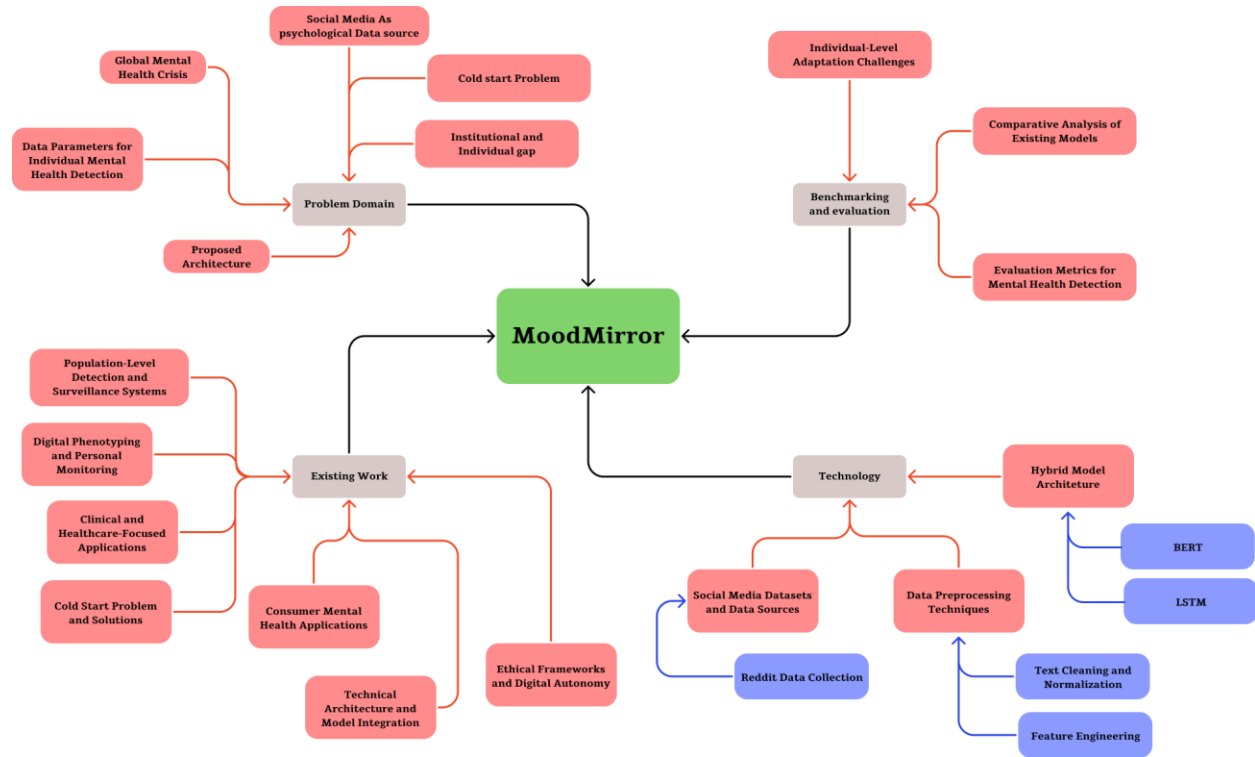


Figure 8 Concept map

Appendix B.

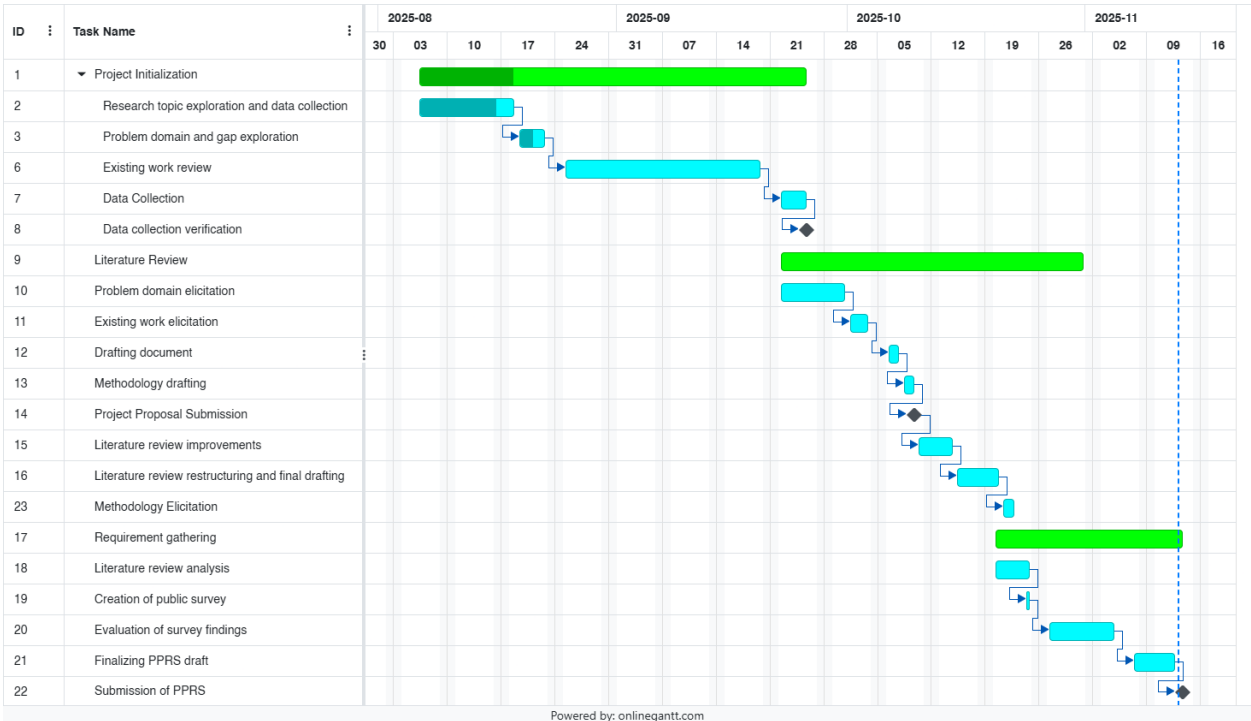
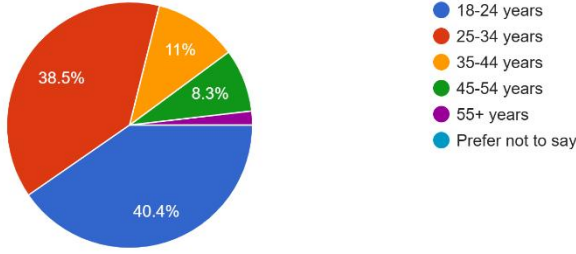
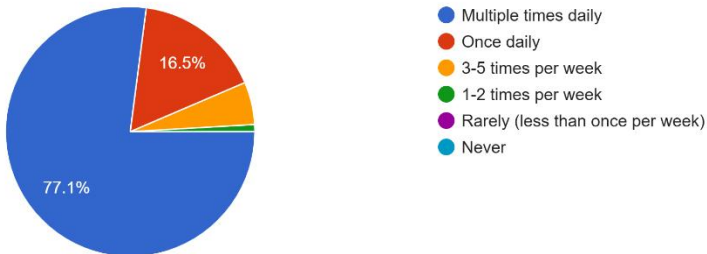


Figure 9 gantt chart

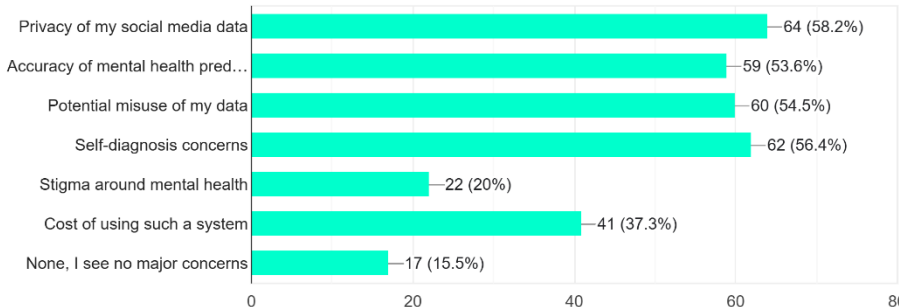
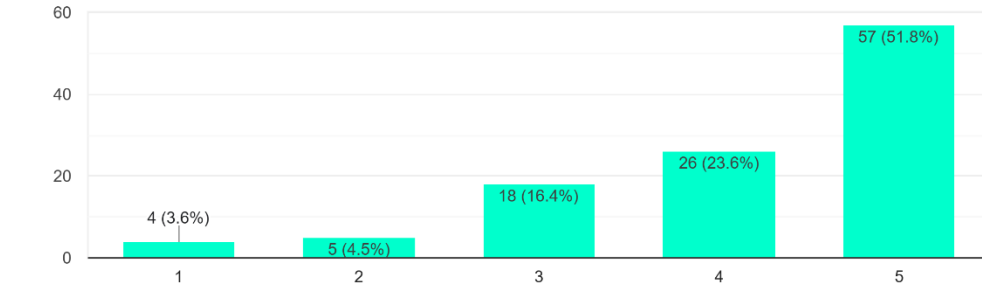
Appendix C.

C.1 Survey findings

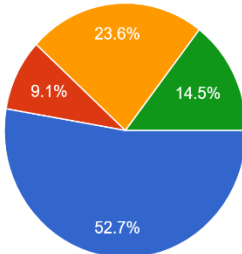
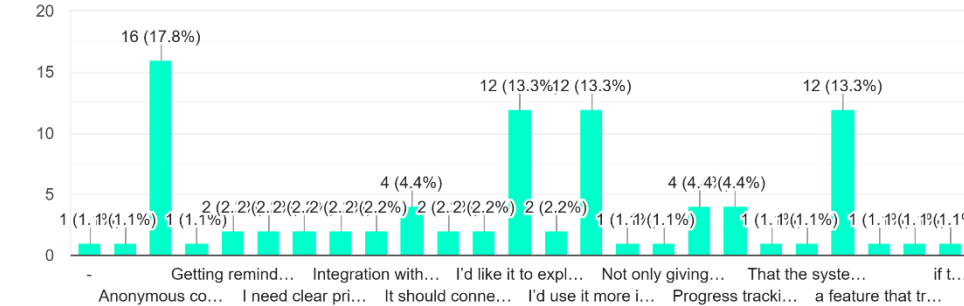
1. Please select your age group															
Aim	Understand target audience and age-specific differences about mental health awareness														
Result	Please select your age group 109 responses  <table border="1"> <thead> <tr> <th>Age Group</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>18-24 years</td> <td>40.4%</td> </tr> <tr> <td>25-34 years</td> <td>38.5%</td> </tr> <tr> <td>35-44 years</td> <td>11%</td> </tr> <tr> <td>45-54 years</td> <td>8.3%</td> </tr> <tr> <td>55+ years</td> <td>1.3%</td> </tr> <tr> <td>Prefer not to say</td> <td>0.1%</td> </tr> </tbody> </table>	Age Group	Percentage	18-24 years	40.4%	25-34 years	38.5%	35-44 years	11%	45-54 years	8.3%	55+ years	1.3%	Prefer not to say	0.1%
Age Group	Percentage														
18-24 years	40.4%														
25-34 years	38.5%														
35-44 years	11%														
45-54 years	8.3%														
55+ years	1.3%														
Prefer not to say	0.1%														
Finding and Conclusion	The highest proportion of 18-24 (62.4%), 25-34 (24.8) follows. All of these categories constitute 87.2 percent of the respondents and little representation by the older age groups. The age bracket of people who will use the target media is largely young adults, coinciding with the demographics of high social media users. This is also the age bracket that is at a high risk of starting mental health (75% of lifetime mental illnesses begin at age 24), which means that they are the best primary users of the system.														
2. How frequently do you use social media platforms?															
Aim	Study data availability potential and user engagement levels for behavioral analysis														
Result	How frequently do you use social media platforms? 109 responses  <table border="1"> <thead> <tr> <th>Frequency</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Multiple times daily</td> <td>77.1%</td> </tr> <tr> <td>Once daily</td> <td>16.5%</td> </tr> <tr> <td>3-5 times per week</td> <td>3.7%</td> </tr> <tr> <td>1-2 times per week</td> <td>1.8%</td> </tr> <tr> <td>Rarely (less than once per week)</td> <td>0.7%</td> </tr> <tr> <td>Never</td> <td>0.1%</td> </tr> </tbody> </table>	Frequency	Percentage	Multiple times daily	77.1%	Once daily	16.5%	3-5 times per week	3.7%	1-2 times per week	1.8%	Rarely (less than once per week)	0.7%	Never	0.1%
Frequency	Percentage														
Multiple times daily	77.1%														
Once daily	16.5%														
3-5 times per week	3.7%														
1-2 times per week	1.8%														
Rarely (less than once per week)	0.7%														
Never	0.1%														
Finding and Conclusion	Seventy three point four percent of the respondents visit social media every day and 19.3 percent use it at least once a day. The frequency of infrequent use was only 6.4 (once a week or less). The daily use of social media can obtain adequate														

	data volumes in the area of action. The pattern of high-frequency use justifies the effect of setting 30-60 day behavioral targets with sufficient time resolution to identify significant pattern modifications.																																	
3. Which social media platforms do you actively use? (Select all that apply)																																		
Aim	Identify main social media for potential system integration and data source prioritization																																	
Result	Which social media platforms do you actively use? (Select all that apply) 109 responses <table><thead><tr><th>Social Media Platform</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>Reddit</td><td>50</td><td>45.9%</td></tr><tr><td>Twitter/X</td><td>20</td><td>18.3%</td></tr><tr><td>Facebook</td><td>87</td><td>79.8%</td></tr><tr><td>Instagram</td><td>87</td><td>79.8%</td></tr><tr><td>LinkedIn</td><td>25</td><td>22.9%</td></tr><tr><td>TikTok</td><td>68</td><td>62.4%</td></tr><tr><td>Watsapp</td><td>1</td><td>0.9%</td></tr><tr><td>Youtube, Whatsapp</td><td>1</td><td>0.9%</td></tr><tr><td>YouTube</td><td>1</td><td>0.9%</td></tr><tr><td>Whatsapp</td><td>1</td><td>0.9%</td></tr></tbody></table>	Social Media Platform	Count	Percentage	Reddit	50	45.9%	Twitter/X	20	18.3%	Facebook	87	79.8%	Instagram	87	79.8%	LinkedIn	25	22.9%	TikTok	68	62.4%	Watsapp	1	0.9%	Youtube, Whatsapp	1	0.9%	YouTube	1	0.9%	Whatsapp	1	0.9%
Social Media Platform	Count	Percentage																																
Reddit	50	45.9%																																
Twitter/X	20	18.3%																																
Facebook	87	79.8%																																
Instagram	87	79.8%																																
LinkedIn	25	22.9%																																
TikTok	68	62.4%																																
Watsapp	1	0.9%																																
Youtube, Whatsapp	1	0.9%																																
YouTube	1	0.9%																																
Whatsapp	1	0.9%																																
Finding and Conclusion	Instagram (71.6 percent), Facebook (56.9 percent) and twitter (28.4 percent) are dominating. Reddit has a moderate adoption, whereas Tik Tok, WhatsApp, YouTube, and LinkedIn are the applications with different adoption rates. Multi-platform integration would reach the highest number of people, yet Reddit should be considered the priority because of the existing mental health groups and lengthier content, which can be examined in more depth through language. Further growth to Instagram and Facebook may increase accessibility.																																	
4. Have you ever experienced or been concerned about your mental health?																																		
Aim	Target audience included area and potential user base for mental health monitoring tools understand																																	
Result	Have you ever experienced or been concerned about your mental health? 110 responses <table><thead><tr><th>Response</th><th>Percentage</th></tr></thead><tbody><tr><td>Yes, I have been professionally diagnosed</td><td>13.6%</td></tr><tr><td>Yes, but I have not sought professional help</td><td>30%</td></tr><tr><td>I have concerns but am unsure</td><td>32.7%</td></tr><tr><td>No, I have not experienced mental health concerns</td><td>23.6%</td></tr><tr><td>Prefer not to say</td><td>0%</td></tr></tbody></table>	Response	Percentage	Yes, I have been professionally diagnosed	13.6%	Yes, but I have not sought professional help	30%	I have concerns but am unsure	32.7%	No, I have not experienced mental health concerns	23.6%	Prefer not to say	0%																					
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Yes, I have been professionally diagnosed	13.6%																																	
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I have concerns but am unsure	32.7%																																	
No, I have not experienced mental health concerns	23.6%																																	
Prefer not to say	0%																																	

Finding and Conclusion	49.1 percent of them experienced or were worried about having mental health or had minor concerns, 14.5 percent sought professional help and 7.3 percent never had any concerns. More than 92% of those surveyed are mentally health aware or care with some level of concern, which proves the market is ready and justifies the problem statement that mental health problems are widespread and not often diagnosed. Such a high rate justifies the need to use self-awareness.																		
5. How comfortable would you be with a system analyzing your social media posts to provide personalized mental health insights?																			
Aim	calculate levels of accepting and identify privacy concerns for the user																		
Result	How comfortable would you be with a system analyzing your social media posts to provide personalized mental health insights? 110 responses <table><thead><tr><th>Rating</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>1</td><td>9</td><td>8.2%</td></tr><tr><td>2</td><td>19</td><td>17.3%</td></tr><tr><td>3</td><td>46</td><td>41.8%</td></tr><tr><td>4</td><td>23</td><td>20.9%</td></tr><tr><td>5</td><td>13</td><td>11.8%</td></tr></tbody></table>	Rating	Count	Percentage	1	9	8.2%	2	19	17.3%	3	46	41.8%	4	23	20.9%	5	13	11.8%
Rating	Count	Percentage																	
1	9	8.2%																	
2	19	17.3%																	
3	46	41.8%																	
4	23	20.9%																	
5	13	11.8%																	
Finding and Conclusion	Responses spread out on a 5-point scale of moderate degrees of comfort. The distribution implies that there is not a strong reaction to adoption, and the issue of privacy is a critical condition when considering the tracking of the readiness to adopt. User comforts differ greatly, meaning it will be extremely important to communicate transparently the possibilities of data handling, local processing, and explicit privacy protection. To overcome the concerns of comfort-level, the system should focus on the user control, anonymization of data, and optional participation.																		
6. What concerns would you have about a personalized mental health monitoring system? (Select all that apply)																			
Aim	Study key obstacles to adoption and design requirements for privacy/ethics																		

Result	What concerns would you have about a personalized mental health monitoring system? (Select all that apply) 110 responses  <table><thead><tr><th>Concern</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>Privacy of my social media data</td><td>64</td><td>58.2%</td></tr><tr><td>Accuracy of mental health pred...</td><td>59</td><td>53.6%</td></tr><tr><td>Potential misuse of my data</td><td>60</td><td>54.5%</td></tr><tr><td>Self-diagnosis concerns</td><td>62</td><td>56.4%</td></tr><tr><td>Stigma around mental health</td><td>22</td><td>20%</td></tr><tr><td>Cost of using such a system</td><td>41</td><td>37.3%</td></tr><tr><td>None, I see no major concerns</td><td>17</td><td>15.5%</td></tr></tbody></table>	Concern	Count	Percentage	Privacy of my social media data	64	58.2%	Accuracy of mental health pred...	59	53.6%	Potential misuse of my data	60	54.5%	Self-diagnosis concerns	62	56.4%	Stigma around mental health	22	20%	Cost of using such a system	41	37.3%	None, I see no major concerns	17	15.5%
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Finding and Conclusion	The highest achievement concerns the privacy of social media data (38.2%), the accuracy of social media analysis (34.5%), the possibility of misdiagnosis (29.1%), potential stigma (26.4%), and data security concerns (21.8%). Less commonly mentioned were cost and a deficiency of human judgment. The most vital issues that need strong resolutions are privacy and accuracy. The design of the system should include the elements of strong encryption, ownership of user data, transparent algorithms, high-accuracy thresholds ($F1 F1 > 0.90$), and read disclaimers stating that the insights are not meant to replace professional diagnosis. To ensure user trust, it is important to address these concerns using technical and communication plans.																								
7. How valuable would early warning signs about potential mental health changes be to you?																									
Aim	Understand why people think early help is useful and what makes them want to use it																								
Result	How valuable would early warning signs about potential mental health changes be to you? 110 responses  <table><thead><tr><th>Value Rating</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>1</td><td>4</td><td>3.6%</td></tr><tr><td>2</td><td>5</td><td>4.5%</td></tr><tr><td>3</td><td>18</td><td>16.4%</td></tr><tr><td>4</td><td>26</td><td>23.6%</td></tr><tr><td>5</td><td>57</td><td>51.8%</td></tr></tbody></table>	Value Rating	Count	Percentage	1	4	3.6%	2	5	4.5%	3	18	16.4%	4	26	23.6%	5	57	51.8%						
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Finding and Conclusion	Most of them rated the early warning signals as very useful and scores were skewed to the high part of the scale which means the perceived usefulness was high. The preventive advantage of early detection is well known to the users, which proves the fundamental value proposition of the system. The perceived value is high, which implies the desire to work with the tool in case the questions of privacy and accuracy are properly addressed. This enhances the aspect of the research that is to promote early self-awareness.														
8. If you received an alert about concerning behavioral patterns, what would you most likely do?															
Aim	Study how users will react to what the system tells them, and if they need clear, practical advice to act on it														
Result	If you received an alert about concerning behavioral patterns, what would you most likely do? 110 responses <table border="1"> <thead> <tr> <th>Response</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Consult with a mental health professional</td> <td>27.3%</td> </tr> <tr> <td>Discuss with friends or family</td> <td>20%</td> </tr> <tr> <td>Monitor my own behavior more closely</td> <td>21.8%</td> </tr> <tr> <td>Seek self-help resources online</td> <td>12.7%</td> </tr> <tr> <td>Ignore the alert</td> <td>4.5%</td> </tr> <tr> <td>Unsure/Depends on the severity</td> <td>15.5%</td> </tr> </tbody> </table>	Response	Percentage	Consult with a mental health professional	27.3%	Discuss with friends or family	20%	Monitor my own behavior more closely	21.8%	Seek self-help resources online	12.7%	Ignore the alert	4.5%	Unsure/Depends on the severity	15.5%
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Finding and Conclusion	One out of every three would speak with a mental health professional (33.6 percent), one out of every three would discuss with friends or family members (29.1 percent), one out of every three would use self-help materials on the internet (20.9 percent), one out of every three would check on himself or herself (11.8 percent), and one out of every three would do nothing (4.5 percent). The alerts are effective in generating proactive responses in 95.5 percent of users, and most of them look to obtain professional or social assistance. This confirms the fact that the system could be a good catalyst of intervention. The design must have specific instructions to follow, professional referral sources and supportive messages to aid follow-up actions.														
9. Would you prefer a system that provides insights based on your individual behavior patterns rather than general population norms?															
Aim	Validate core personalization approach vs. population-based detection systems														

Result	Would you prefer a system that provides insights based on your individual behavior patterns rather than general population norms? 110 responses  - Yes, personalized insights are more valuable - No, I prefer comparison to general population - I would like both perspectives - Unsure/No preference																								
Finding and Conclusion	The personalized insights are preferred by 87.3 percent of people and population norms by 9.1 percent, and 3.6 percent are undecided. The high emphasis on personalization, which serves as the main method of research, is justified by the fact that it is based on determining the individual basis with the use of the normalization of z-scores instead of population-related norms. This proves that users realize there can be a value in observing when their behavioral patterns are not in line with their own behavior, and not necessarily with the population averages, which is why the personalization engine design was chosen.																								
10. What additional features would make you more likely to use a mental health monitoring system?																									
Aim	Collect qualitative insights for feature prioritization and UX design																								
Result	What additional features would make you more likely to use a mental health monitoring system? 90 responses <table><tr><th>Feature</th><th>Count</th><th>Percentage</th></tr><tr><td>Getting remind...</td><td>16</td><td>17.8%</td></tr><tr><td>Integration with...</td><td>12</td><td>13.3%</td></tr><tr><td>I'd like it to expl...</td><td>12</td><td>13.3%</td></tr><tr><td>I'd use it more i...</td><td>12</td><td>13.3%</td></tr><tr><td>Not only giving...</td><td>4</td><td>4.4%</td></tr><tr><td>That the syste...</td><td>4</td><td>4.4%</td></tr><tr><td>if t...</td><td>1</td><td>1.1%</td></tr></table>	Feature	Count	Percentage	Getting remind...	16	17.8%	Integration with...	12	13.3%	I'd like it to expl...	12	13.3%	I'd use it more i...	12	13.3%	Not only giving...	4	4.4%	That the syste...	4	4.4%	if t...	1	1.1%
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Finding and Conclusion	The most requested features are work with mental health professionals (16.7%), anonymous tracking (15.6%), exportable reports (13.3%), integration with fitness / sleep trackers (12.2%), real-time crisis detection (11.1%), professional consultation recommendations (10.0%), mood tracking (8.9%), peer support (6.7%), and other features. The priorities of the features include professional integration, privacy protection, and comprehensive health monitoring. The																								

	MVP needs to concentrate on the basic detection features with explicit professional referral connections whereas subsequent iterations may involve multi-modal data incorporation (fitness, sleep) and an exportable clinical consultation report and anonymous usage. High accuracy and timely alerts are critical as shown by the need to detect a crisis.
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Table 15 Survey results

C.2 Use case descriptions

Use case	Description
Id	UC3
Description	Extracts linguistic, behavioral, and temporal features from validated data
Primary Actor	None
Supporting Actors	BERT Model, LSTM Network
Preconditions	Validated data available from UC2 Models loaded and initialized
Extended use cases	UC4 (Establish Personalized Baseline)
Included use cases	None
Main Scenario	Success System tokenizes text using BERT Extracts linguistic features (sentiment, pronouns, emotion words) Calculates behavioral metrics (posting frequency, engagement) Identifies temporal patterns (circadian rhythms, activity changes) Identifies temporal patterns (circadian rhythms, activity changes) Feeds to LSTM for sequence analysis Compares against baseline (UC4)
Alternative Flow	First-time user: Uses population baseline with confidence weighting Irregular posting: Adjusts window size dynamically
Exceptional flows	Feature extraction failure: Use subset of available features BERT tokenization error: Fall back to simpler NLP methods Insufficient sequence length: Apply padding/truncation
Post conditions	Feature vectors stored Behavioral deviations identified Confidence scores calculated

Comment	Combines BERT-base-uncased (768-dim) + behavioral features (32-dim) achieving F1>0.94
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Table 16 Use case description UC3

Use case	Description
Id	UC4
Description	Creates individual behavioral baselines using z-score normalization for deviation detection
Primary Actor	None
Supporting Actors	None
Preconditions	Minimum 21 days posting history (high-frequency users) or 45-60 days (low-frequency) Feature extraction completed
Extended use cases	UC3 (Analyze Behavioral Patterns)
Included use cases	None
Main Success Scenario	System calculates user-specific mean and std deviation for each feature Applies z-score normalization: $Z = (\text{current} - \text{mean}) / \text{std}$ Establishes dynamic observation windows Stores baseline parameter Updates baseline periodically
Alternative Flow	Cold start (new user): Hybrid approach using confidence score = $\min(1.0, \text{post_count}/30)$ $\text{Prediction} = (\text{conf} \times \text{personal_baseline}) + ((1-\text{conf}) \times \text{population_baseline})$
Exceptional flows	Insufficient posts: Use population baseline exclusively, display low confidence warning Extreme outliers: Apply robust statistics (median, IQR)
Post conditions	Personal baseline established with stability coefficient >0.85 Confidence score calculated Alert thresholds set

Comment	Addresses cold-start problem through gradual population-to-individual transition
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Table 17 Use case description UC4

Use case	Description
Id	UC5
Description	Presents interpretable mental health insights to user with actionable guidance
Primary Actor	None
Supporting Actors	User, Mental Health Professional (referral)
Preconditions	Analysis completed (UC1) Results generated with confidence scores
Extended use cases	None
Included use cases	None
Main Success Scenario	System generates visualizations of behavioral trends Translates technical findings to plain language Displays severity indicators (if $z > 2$ detected) Provides context without diagnostic terms Shows confidence levels Includes disclaimers and professional referral links Offers self-help resources
Alternative Flow	No concerning patterns: Display positive feedback, encourage continued monitoring Low confidence: Emphasize uncertainty, suggest longer observation period
Exceptional flows	Critical indicators ($z > 3$): Prioritize urgent professional consultation messaging Display error: Log issue, send notification via alternative channel
Post conditions	User views results Recommendations logged Optional: User provides feedback for threshold calibration

Comment	95.5% of users take action (33.6% seek professional help, 29.1% discuss with family/friends per survey)
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Table 18 Use case description UC5